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Macroevolutionary divergence along allometric lines of least resistance in frog hindlimb traits and its effect on locomotor evolution

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Running head: Frog jumping performance and constraints

MS type: Article

Keywords: Anura; functional performance; genetic constraints; macroevolution; microevolution; morphological evolution

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Abstract

Understanding whether and why microevolutionary patterns of trait covariation match macroevolutionary divergence is essential for linking evolution at different timescales. However, recent work has focused on developmental constraints for alignment between intraspecific variation and divergence, neglecting a potential role of natural selection on function to connect these scales. Here, we compare the support for the selection and constraint hypotheses to explain both phenotypic trait covariation and species divergence. To test these hypotheses, we collected data on hindlimb and jumping performance traits within and across species of two frog genera. We compared patterns of within-species phenotypic variation (the P-matrix) with divergence and selective covariance matrices, from which we could extract the major axes of the realized adaptive landscape (AL), the directions in which adaptive peaks shifted the most over evolutionary time. We also tested whether the major axes of the AL were related to selection on jumping performance. We found high alignment between patterns of variation across scales. Most divergence occurred in allometric size, defined as the first eigenvector of the P-matrix. However, jumping performance gradients were unaligned with the major axes of the AL and the P-matrix. Across species, however, evolution of maximum acceleration showed a strong negative relationship with changes in allometric size. We infer that the jumping peak evolved under fluctuating selection, and species have tracked the peak along the direction of most within-species variation, allometric size. We conclude that long-term hindlimb divergence was constrained by developmental interactions among traits associated with growth and not net directional selection. Nonetheless, divergence on size indirectly influenced jumping evolution.

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Introduction

Most phenotypes affecting fitness are complex, composed of several traits and influenced by both genetic and environmental factors (Lande 1979; Lande and Arnold 1983; Wagner and Altenberg 1996; Lynch and Walsh 1998). Although the short-term evolution of complex phenotypes is well characterized (but sometimes hard to predict), understanding the evolution of complex phenotypes at larger evolutionary timescales is challenging. Connecting microevolution to macroevolutionary divergence is complicated by the evolution of the genetic architecture of traits, in which trait covariances are widespread in complex phenotypes. Genetic architecture can remain stable or change over long periods of time (Arnold et al. 2008), with each outcome affecting phenotypic divergence differently. Unraveling which evolutionary processes shape genetic covariances and correlations over long timescales is essential to better understand the evolution of complex phenotypes, and ultimately, to better connect micro- and macroevolution.

At microevolutionary scales, the divergence of complex phenotypes depends on the genetic architecture of multiple traits, represented by the additive genetic variance-covariance matrix (the G-matrix) (Lande 1979; Arnold et al. 2001, 2008). The G-matrix is composed of genetic variation of traits on its diagonal and genetic covariation between traits in the off-diagonal, reflecting the co-inheritance of multiple traits (Lande 1979; Arnold 1992; Lynch and Walsh 1998). Higher genetic covariation between traits may bias evolutionary trajectories of phenotypic means away from the direction of selection, toward directions that have most genetic variation (Lande 1979; Lande and Arnold 1983; Schluter 1996; Hansen and Houle 2008; Walsh and Blows 2009). Yet, given enough time, phenotypic means may reach adaptive optima and the G-matrix would no longer affect the evolution of phenotypic means (Arnold et al. 2001).

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However, genetic architecture can evolve over long timescales, making its role in shaping long-term phenotypic evolution less straightforward than at microevolutionary scales (Turelli 1988; Steppan et al. 2002; Arnold et al. 2008). Both theoretical (Jones et al. 2004, 2012, 2014; Revell 2007; Arnold et al. 2008) and empirical (Cano et al. 2004; Eroukhmanoff and Svensson 2011; Björklund et al. 2013; Careau et al. 2015; Renaud et al. 2021; Chantepie et al. 2024) studies have shown that the G-matrix can change, although the changes may be subtle even over millions of years (McGlothlin et al. 2022). On the other hand, several processes can promote G-matrix stability, such as a strong correlated mutational effects (Lande 1979; Hohenlohe and Arnold 2008), persistent correlational selection molding genetic correlations (Cheverud 1984; Jones et al. 2003; Revell 2007), and persistent directional selection that aligns the G-matrix and the adaptive landscape (Jones et al. 2004, 2007, 2014; Pavlicev et al. 2011; Melo and Marroig 2015; Houle et al. 2017). Therefore, connecting the evolution of the G-matrix with phenotypic divergence at the macroevolutionary scale is difficult and depends on the relative strengths of distinct evolutionary forces shaping the G-matrix and the species' phenotypic means.

Empirical evidence of species divergence aligned with the axis of most genetic variation (also known as g_{max}) was first shown by Schluter (1996) across different vertebrates. Yet divergence was less aligned with g_{max} in distantly related species (around 4 million years of divergence) compared with species that diverged more recently (Schluter 1996). Subsequent work also showed a relatively short-term effect of genetic constraints on divergence in different systems (Bégin and Roff 2004; McGuigan et al. 2005; Hansen and Houle 2008; Chenoweth et al. 2010; Bolstad et al. 2014; Opedal et al. 2022, 2023; Holstad et al. 2024). However, recent papers have shown that such effects can persist much longer than previously thought, spanning tens of millions of years (Houle et al. 2017; McGlothlin et al. 2018). More extensive evidence shows

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long-term divergence aligned with the axis of most phenotypic variation (p_{max}), which is frequently aligned with g_{max} (Marroig and Cheverud 2005, 2010; Renaud et al. 2006; Hunt 2007; Simon et al. 2016; Machado 2020; Mongle et al. 2022). Such evidence suggests that the G-matrix preserves its structure and its influence on phenotypic divergence over long periods of time (Arnold et al. 2001; McGlothlin et al. 2018).

What evolutionary scenarios might explain such a persistent relation between the G-matrix and divergence? First, the constraint hypothesis states that a persistent correlation of mutational effects (pleiotropic mutations expressed on the mutational matrix, the M-matrix) promotes stability of genetic correlations (Lande 1980; Jones et al. 2003; Houle et al. 2017), biasing most divergence along axes of most genetic variation, whether divergence was mainly caused by random drift or selection. If selection has been the main driver of divergence, the persistent effect of constraints may indicate the existence of an adaptive landscape with multiple adaptive peaks or with a fluctuating peak, in which peaks are easier to reach in directions of most genetic variance or lineages chase the peak along directions of most variance, respectively (Steppan et al. 2002; Melo et al. 2016b; Holstad et al. 2024). Second, the selection hypothesis implicates evolution along selective lines of least resistance (SLLR), which represent ridges in the adaptive landscape, that is, directions under weakest stabilizing selection, along which changes in phenotype will produce little or no change in average fitness (Arnold et al. 2001; Hohenlohe and Arnold 2008; Arnold 2023). SLLR can be estimated as the last eigenvector of a matrix of multivariate stabilizing selection coefficients (the γ-matrix), which represents the curvature and orientation of the adaptive landscape (AL; Lande and Arnold 1983, Phillips and Arnold 1989; Arnold et al. 2001). The SLLR would then favor peak shifts in their own direction

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to shape phenotypic means, while also aligning the G-matrix with the adaptive landscape (Arnold et al. 2001; Jones et al. 2003, 2012; Arnold 2023).

All of these scenarios are plausible, yet those that include selection have been most supported in the literature. Such supporting studies have either shown phenotypic divergence to be driven by selection (Marroig and Cheverud 2005, 2010; Simon et al. 2016; Machado 2020) or have been done with morphological traits that had clear functional relevance, implying diversifying selection (e.g., Renaud et al. 2006; Kolbe et al. 2011; McGlothlin et al. 2018; Mongle et al. 2022; Opedal et al. 2022). Moreover, most of these studies considered feeding or locomotor traits, suggesting a role of selection on function mediating the effects of constraints on divergence. Long-term correlational selection is expected to shape patterns of trait correlations to maintain functional integration, that is, to maintain higher correlations among traits that act together to perform a specific function (Olson and Miller 1958; Cheverud 1984; Estes and Arnold 2007). Thus, selection on functional performance may align long-term phenotypic divergence with intraspecific variation by shaping both trait correlations and phenotypic means.

However, functional performance is rarely incorporated in studies aiming to link micro- to macroevolution. Most studies that compare both scales have focused on scaling relationships (effects of changes in size on performance; e.g., Toro et al. 2003) or considered few species (e.g., Taverne et al. 2021). Moreover, recent work has focused on developmental reasons for alignment between intraspecific variation and interspecific divergence (e.g., Mongle et al. 2022, Machado et al. 2023). Finally, no empirical work has addressed how selection on functional performance may shape both trait correlations and phenotypic divergence, precluding a test of selection on function as a mechanism to explain the lasting effect of genetic constraints on divergence.

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In this paper, we pursue a deeper understanding of the link between genetic architecture within populations and phenotypic divergence across species by directly considering selection on trait function. We studied 60 species of frogs belonging to two genera, one arboreal (*Boana*) and the other terrestrial (*Leptodactylus*). We collected morphological data (on hindlimb traits) for all species and jumping performance data (maximum acceleration and velocity) for 18 of those species (Figure 1). We also collected intraspecific data to estimate phenotypic within-species patterns (P-matrices), which allowed us to test whether within-species trait covariation patterns matched between-species divergence and whether divergence in both morphology and performance were driven by selection. Although we could not estimate a jumping performance γ-matrix and extract SLLR, we estimated the major axes of the AL shaping hindlimb morphology by estimating a selective covariance matrix, the covariance matrix of peak movements over evolutionary time (Felsenstein 1988; Marroig and Cheverud 2010; Machado 2020). Then, we estimated functional performance gradients—the effects of hindlimb morphology on jumping performance—and tested how well they aligned with phenotypic lines of least resistance and with major axes of the AL. If functional performance aligned with both within-species patterns of variation and with the AL, we would support the selection hypothesis and infer that selection on function mediates the alignment of micro- and macroevolution. If functional performance aligned with neither within-species patterns nor with the AL, we would support the constraint hypothesis and infer that constraints unrelated to function drove the alignment between micro- and macroevolution.

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Methods

Specimens, sample sizes, and taxonomy

We caught individuals in the wild from seven species of *Boana* and eight of *Leptodactylus* in French Guiana in 2013 and 2015. We generally targeted males to reduce potential effects of sexual dimorphism (see below) and because males were calling. In *Boana*, mature males could be identified by the presence of prepollical spines (Pinheiro et al. 2022). However, some species of *Leptodactylus* did not show clear sexual dimorphism, so we verified sex via dissection after performance-data collection (see below). We sampled a total of 131 individuals across both genera, collecting 1–17 individuals per species (mean n = 7.6) for those we only sampled for interspecific analyses (Table S1). To improve estimates of within-species trait covariation patterns, we more extensively sampled the most abundant species in our field sites for each genus: n = 48 for *Leptodactylus mystaceus* and n = 29 for *Boana boans*. These samples represent data for both morphology and jumping performance.

To better estimate morphological trait covariation within species, we measured museum specimens collected from localities close to French Guiana, so that final sample sizes were n = 53 for *L. mystaceus* and n = 45 for *B. boans* for morphology-only analysis. Likewise, we included data from the literature (Moen et al. 2013, 2016) and collected new morphological data from additional museum specimens (19 species of *Boana* and 16 species of *Leptodactylus).* In sum, for morphological analysis we analyzed 479 individuals across 33 species of *Boana* and 27 of *Leptodactylus*. For morphological and performance analysis, we had 167 individuals representing nine *Boana* and nine *Leptodactylus* species (Figure 1). See Table S1 for the sample size for each species and associated numbers of males and females. Finally, we confirmed or updated species names based on collection localities and recent taxonomic literature (see Supplementary Data for details).

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Jumping performance

We measured jumping forces using a Kistler piezoelectric force plate (Squirrel, model Z17097) and charge amplifier (model 9865), collecting data at 500 Hz. Body mass was measured before jumping trials on an Ohaus balance precise to 0.1g. Each frog was then stimulated to jump from the force plate repeatedly until at least three strong jumps were recorded per individual, with one day of rest between trials. To extract maximum jumping forces, we first isolated the highest force peaks in the vertical direction in each force trace. We then extracted the corresponding horizontal (X, Y) and vertical (Z) forces as text files using BioWare software (version 5.4.8.0, Kistler).

We describe our analysis of force profiles in detail in Online Supplement and Figures S1 and S2. In brief, we used a custom MATLAB script (version R2020a, MathWorks; script developed by Craig McGowan) to smooth, center, and integrate the X, Y, and Z forces traces independently, then calculated single-dimensional jumping velocity and acceleration profiles (Schwaner et al. 2021). We then extracted the peak acceleration and velocity for each trace for each individual as our two performance variables. We used the peak values from the single best jump, defined as the jump with maximum acceleration, as data for each individual (Wilson et al. 2000; James and Wilson 2008). The force trace data for each jumping peak and the MATLAB script are provided in Dryad and Zenodo (Simon et al. 2025). All subsequent analyses were performed in the R programming environment (version 4.2.2, R Core Team 2022) and can be reproduced using the data file and supplementary R scripts associated with this paper (Simon et al. 2025).

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Morphology

After performance trials, we euthanized individuals, preserved them in formalin, then stored them in 70% ethanol. We primarily collected data on hindlimb traits because hindlimb length and muscle mass strongly predict frog jumping performance within species (e.g., Emerson 1978; Wilson et al. 2000; James et al. 2005, 2007) and especially across species (Gomes et al. 2009; Moen et al. 2013, 2021; Citadini et al. 2018; Moen 2019; Juarez et al. 2020). We used a digital caliper (to 0.1mm) to externally measure snout-to-vent length (SVL), femur length, tibiofibula length, tarsus length, foot length, the largest width of the upper and lower leg, and the width perpendicular to the last two measurements (Juarez et al. 2023; Morinaga et al. 2023). Using the leg widths as axes of an ellipse to calculate area, we estimated the upper- and lower-leg cross-sectional areas (CSA), which should be proportional to muscle CSA, a key component of jumping performance in anurans (Hellam and Podolsky 1969; Peplowski and Marsh 1997; Olson and Marsh 1998; Astley 2016). We took the square root of the muscle CSA to put it on the same linear scale as our other measurements. Moreover, to be able to interpret morphological patterns in an allometric context (given that size is an important axis of variation in all species; see Results), we natural-log-transformed all trait values.

Within-species phenotypic matrix and matrix similarity

To characterize within-species trait covariation, we estimated three within-species variance-covariance matrices (P-matrices) for the species with highest sample sizes: *Boana boans*, *Boana raniceps*, and *Leptodactylus mystaceus* for morphology, with sample sizes of 45, 44, and 53, respectively. We first tested for effects of sex within each species on the six hindlimb traits. We used a multivariate linear regression model with all traits as the dependent variables and only sex

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as the fixed factor for *B. boans* and *L. mystaceus* (*B. raniceps* had no females). We then used the residuals of the linear models to estimate the P-matrices in the function "CalculateMatrix" (*evolqg* package, version 3.1; Melo et al. 2016a).

To determine if patterns of within-species trait covariation were shared by the three species, we compared P-matrices using two matrix similarity indices. First, random skewers (Cheverud and Marroig 2007) give the similarity of a pair of matrices as the average vector correlation of response vectors obtained from applying random selection vectors. Second, the Krzanowski projection (Krzanowski 1979) gives similarity in the shared eigenvector subspace (three pairs of eigenvectors) between the matrices being compared. We provide full details of these methods in the Online Supplement. Both indices showed high similarity among all three P-matrices (above 0.89 for all pairwise comparisons; Table S2). Due to this high similarity, we estimated pooled P-matrices for each genus as our best estimate of within-species patterns. By pooling all species from the same genus, we achieved higher sample sizes to estimate six by six intraspecific P-matrices: n = 242 for *Boana* and n = 215 for *Leptodactylus*. We used multivariate linear regression models with both species and sex as fixed factors, as well as their interaction, and then used the residuals of the models to estimate the pooled P-matrices for *Boana* and *Leptodactylus* using the function "CalculateMatrix" from *evolqg*. However, we also used the P-matrices from our three species with high intraspecific samples in some analyses to test the sensitivity of our results to using pooled matrices (see below).

After estimating the pooled P-matrices, we performed an eigenanalysis to interpret which combinations of traits were associated with the most phenotypic variation. We focused on only the first two eigenvectors because they represented over 85% of all trait variation in the three species. We used those first two eigenvectors of the pooled P-matrices as surrogates for genetic

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lines of least resistance, because G-matrices have never been estimated for our study taxa and traits. For morphological traits, P-matrices are good approximations of G-matrices in a diversity of biological organisms, for five reasons. First, morphological traits show moderate to high heritability (Phillips and Arnold 1989; Roff 1996; Porto et al. 2009; Styga et al. 2019). Second, phenotypic and genetic variation of log-transformed traits tend to scale isometrically for several organisms (Holstad et al. 2024). Third, a high similarity between P-matrices of distantly related species, as we found between *B. boans* and *L. mystaceus*, suggests high stability of trait covariation patterns over evolutionary time, compatible with stable G-matrices (Phillips and Arnold 1999; Marroig and Cheverud 2001). Fouth, a single point estimate of the G-matrix may be less predictive of divergence than a pooled P-matrix that represents a long-term average of covariation patterns (Mongle et al. 2022). Finally, traits closely associated with functional performance are likely under selection to favor developmental architectures that do not disrupt functional integration, channeling environmental variation to match genetic patterns.

Divergence variance-covariance matrix and alignment with within-species covariation patterns

We estimated the divergence morphological matrix (D-matrix) for each genus accounting for phylogeny. We used species' means as the data and the anuran phylogeny of Portik et al. (2023). This phylogeny was estimated with phylogenomic analysis on over 300 markers, is the most recent comprehensive phylogeny available, and has branch lengths in units of time. Most of our sampled species were in the phylogeny (see Online Supplement and Supplementary data for taxon-substitution details). We then pruned this phylogeny to contain only our sampled species of *Boana* (n = 33 species) or *Leptodactylus* (n = 27 species).

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We calculated the D-matrix under a Brownian Motion (BM) model of evolution (Lande 1979; Revell 2007; Simon et al. 2016; Machado 2020). The mean rate of evolution of each trait was calculated as the mean square of phylogenetic independent contrasts (PICs, Felsenstein 1985), whereas the mean cross-product of the PICs for two traits was proportional to their mean rate of co-evolution (Revell et al. 2007). We used the function 'ratematrix' in the R package *geiger* (version 2.0.11, Harmon et al. 2008) to estimate the D-matrix for each genus. We also calculated a D-matrix assuming a single-peak Ornstein-Uhlenbeck model of evolution. D-matrices assuming BM and OU were highly similar to each other within and across both genera (see Online Supplement for details). Thus, we chose to use D-matrices under BM because it is the simplest model.

We conducted an eigenanalysis to examine patterns of most divergence in trait combinations and also to test whether directions of most divergence were aligned with directions of most intraspecific variation. We compared the D-matrices with pooled P-matrices, but we also compared them with the three within-species P-matrices. We used the KrzCor method to test for similarity between P-matrices and D-matrices using four pairs of eigenvectors; we excluded the last two eigenvectors because KrzCor will give a similarity of one if all eigenvectors are included (see Online Supplement). Additionally, to check specifically for correlations between the first two eigenvectors of the D-matrix (which corresponded to 98–99% of the variation across species) with the first two eigenvectors of within-species P-matrices and the pooled P-matrix, we also quantified their alignment using vector correlations (the absolute value of the dot product between two normalized vectors). We calculated such correlations separately for each genus. To account for estimation uncertainty for both matrices in a given genus-specific comparison, we constructed 95% confidence intervals (95% CI) for the vector correlations using bootstrapping

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resampling (see Online Supplement for details). Given that vector correlations may be high by chance, we compared these 95% CIs with a null distribution of vector correlations constructed with 1,000 unit-length random 6-element vectors drawn from a random normal distribution of mean zero and unit SD. We considered observed vector correlations as significant when the 95% CI were above the upper 5% tail of the null distribution (above 0.76). We performed all analyses with ln-transformed traits to guarantee covariances were scale-invariant and mean-independent (Bookstein et al. 1985; McGuigan et al. 2005).

Random drift test

To assess which evolutionary process most likely caused divergence in hindlimb traits, we first tested whether species divergence was consistent with genetic drift (Lande 1976, 1979; Lovsfold 1988). More variable traits within an ancestral population evolving by genetic drift will lead to more divergence in species' means in those same traits compared with less variable traits. Thus, modeling the divergence of population means under genetic drift is a function of the ancestral G-matrix, the effective population size (N_e), and the time since divergence (t; Lande 1979; Lovsfold 1988):

$$\boldsymbol{D} = \boldsymbol{G}\left(\frac{t}{N_e}\right) \tag{1}$$

Given that t and N_e are constants, we expect the D-matrix to be proportional to the G-matrix under drift. For morphological data, it is reasonable to assume that the G-matrix can be represented by the pooled P-matrix (Ackermann and Cheverud 2002), so we tested this proportionality expectation using the variance in the direction of eigenvectors of the pooled P-matrix and the variation in species' scores projected onto the same eigenvectors (representing the D-matrix). We also performed drift tests using within-species P-matrices instead of the pooled P-

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matrix. On a log scale, we may then test for proportionality of within- and between-species variation by calculating the regression slope (β) that relates their variances along the eigenvectors of the pooled P-matrix or within-species P-matrix (Ackermann and Cheverud 2002). Under drift, we expect $\beta = 1$. In contrast, if the slope significantly deviates from 1, we reject drift and infer that some form of selection would have to have either increased (e.g., diversifying) or reduced (e.g., stabilizing) divergence relative to the expectation under drift (Arnold et al. 2001). We took phylogeny into account by using phylogenetic independent contrasts (PICs) of the species means, which we projected onto the eigenvectors of the pooled P-matrix or of the species-specific P-matrix. To see further details on this method, see the Online Supplement.

Major axes of the macroevolutionary adaptive landscape

Even if within- and between-species patterns were aligned and random drift were rejected, the divergence across species could still reflect two distinct scenarios: genetic constraints biasing the response to selection (constraint hypothesis) or selection shaping both trait covariance patterns and differences in species' means, aligning the P-matrix with the AL (selection hypothesis). To infer genetic constraints, the first two eigenvectors of the pooled P-matrix (p_{max} and p_2), representing evolutionary lines of least resistance, must not align with the major axes of the AL, representing the directions of most movement of adaptive peaks. Alternatively, if both the pooled P-matrix and the D-matrix were aligned with the AL, it would indicate that trait divergence patterns were produced by the covariance of movement of morphological adaptive peaks. If we find support for the latter scenario, the major axes of the AL could in principle coincide with SLLR. However, we could not test this hypothesis because we would need many more

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individuals within species (at least n = 500) than our highest sample size (n = 53) to properly estimate a populational γ-matrix and extract its last eigenvectors to represent SLLR (Simon et al. 2022).

To estimate the major axes of the AL, we first calculated the selection covariance matrix (the W-matrix), the covariance matrix of directional net-selection gradients, reflecting the covariance of selective pressures acting on multiple traits and producing covariances in trait changes across species (Felsenstein 1988; Marroig and Cheverud 2010; Machado 2020):

$$\boldsymbol{W} = \boldsymbol{G}^{-1}\boldsymbol{D}\,\boldsymbol{G}^{-1} \tag{2}$$

in which $\boldsymbol{G}^{-1}$ is the inverse of the G-matrix and $\boldsymbol{D}$ is the divergence matrix, the variance-covariance matrix of evolutionary changes in species' means. Notice that we use the same matrix parameters as the drift test described above, but now assume a multivariate directional selection model interacting with genetic constraints, and no stabilizing selection (model II in Zeng 1988). This model, $\boldsymbol{D} = \boldsymbol{G}\,\boldsymbol{W}\,\boldsymbol{G}$, describes how independent populations can show correlated evolutionary changes in the D-matrix due to shared constraints (represented by the G-matrix) and similar directional selection pressures (represented by the W-matrix), or both (Felsenstein 1988). It is important to realize that the D-matrix calculated with PICs is equivalent to the average cross-products of $\Delta\mathbf{z}$ from ancestor-descendent lineages, and thus, the W-matrix is a covariance matrix of net-selection gradients (i.e., minimum amount of directional selection required to produce differences in species' means; Felsenstein 1988; Arnold et al. 2001; Machado 2020). Therefore, the eigenvectors of the W-matrix represent the major axes of the realized AL, the main directions of shifts in adaptive peaks. To estimate the W-matrix, we used the inverse of the pooled P-matrix as a surrogate of the G-matrix, and the D-matrix representing between-species patterns, for each frog genus. We scaled both $\boldsymbol{G}$ and $\boldsymbol{D}$ with their respective traces (the sum of

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their diagonals; Bookstein and Mitteroecker 2014) to obtain reasonable magnitudes of variation and covariation of net-selection gradients. This procedure does not change the eigenvectors of the W-matrix.

We then performed an eigen decomposition of the W-matrix to obtain the major axes of the AL (w_{max} and w_2), then correlated them with the eigenvectors of the pooled P-matrix and of the D-matrix. Once again, to account for uncertainty in estimating these matrices, we used the 1,000 bootstrapped pooled P-matrices and D-matrices, to calculate resampled W-matrices and compute the 95% CI for the correlations. A vector correlation was considered significant if the 95% CI was above the 5% upper tail of the null distribution of 6-element vector correlations (0.76). However, we interpreted these vector correlations as simple quantification of the match among directions of the AL, divergence, and constraints, rather than strict statistical tests.

Selection on jumping performance

To infer whether selection on jumping performance relates to the hindlimb divergence across species, we first tested whether jumping performance showed evidence of adaptive evolution. We aimed to test if performance showed an optimal state predicted by species variation in the axis of most divergence across both *Leptodactylus* and *Boana* (i.e., the first axis of the D-matrices, which we inferred to reflect size variation; see Results). In this case, performance would be evolving under an Ornstein-Uhlenbeck (OU) process, in which adaptation is not necessarily instantaneous. In contrast, the predictor—here, species means projected on d_{max}—would be evolving under a BM model (the OUBM model; Hansen et al. 2008). We can interpret the OU process as describing the movement of the adaptive optima of jumping performance over

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evolutionary time in response to changes in morphology, while accounting for constraints and lag in adaptation (Hansen et al. 2008; Hansen 2012).

We compared evolutionary models separately for each performance variable using the package *slouch* (version 2.1.4, Hansen et al. 2008), which allows OU models with continuous predictors. We tested models with d_{max} as the predictor, assuming either a BM model with a trend or an OUBM model. We compared these models to an intercept-only BM, an OU model with a single regime for all species, and an OU model with two regimes (arboreal or terrestrial microhabitat). For the latter, we set internal ancestral states for model-fitting by using ML and an equal-rates model, which showed the best fit to the data, using the function 'ace' (*ape* R package, version 5.7-1, Paradis 2009). We also performed a multivariate OU analysis with both performance variables as response variables. However, our results seemed unreliable (see Online Supplement), so for our data we favor the single-variable models described above.

Prior to analysis, we logged both performance variables and the predictor because we tested for an adaptive allometric relationship among them. To account for sampling error when estimating the regression slopes, we used squared standard errors of both jumping performance variables and species scores on d_{max}, following Ives et al. (2007; Appendix 3). We then used AICc for model comparison, using AICc weights to check the relative support for each model. To be conservative, we interpreted parameters for all models that showed AICc weights above 0.2, indicating modest to high support (Burnham and Anderson 2004).

Functional performance gradients

To further test the relationship between selection on jumping performance and both the pattern of trait covariation and the realized AL, we estimated jumping performance gradients and their

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correlation with p_{max} and p_2, and w_{max} and w_2, respectively. Linear performance gradients are partial linear regression coefficients that reflect the effects of morphology on performance and can be interpreted as part of the total selection on morphology (Arnold 1983). To estimate performance gradients within species, we fit multiple linear regressions of maximum acceleration and velocity on the hindlimb traits, for both *Boana boans* and *Leptodactylus mystaceus*. After calculating the jumping performance gradients, we normalized them to a length of one. We then tested their vector correlation with w_{max} and w_2, and with p_{max} and p_2. We calculated 95% CI for these vector correlations using the resampled distributions of the pooled P-matrix, divergence matrix, and W-matrix, from which we extracted eigenvectors. We performed the same analysis for between-species performance gradients, which quantified the hindlimb traits that most contributed to differences in jumping across species (see Online Supplement). Comparing between-species performance gradients with w_{max} and w_2 may be more relevant (than within-species analysis) to test whether shifts in morphological adaptive peaks relate to species' differences in jumping performance.

We first conducted these analyses using all six hindlimb traits for morphology. However, such gradients may have been poorly estimated with the sample sizes we had within species, and they were hard to interpret functionally. Thus, we also estimated performance gradients using just two morphological traits: leg length (the sum of the four bone lengths) and the square root of the total muscle CSA (the sum of the two muscle measurements). This simpler analysis made performance gradients more reliable and easier to interpret, while still retaining the bone length by muscle CSA contrast found as relevant in the species morphological divergence (see Results). We also estimated pooled P-, divergence-, and W-matrices with just leg length and muscle CSA to compare with the simplified performance gradients.

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Results

Patterns of trait covariation are shared within and between species

The patterns of trait covariation were similar across all within-species P-matrices, and accordingly, also between pooled P-matrices for each genus (Table S2). More importantly, trait covariation patterns represented by either pooled P-matrices or within-species P-matrices were moderately to highly similar to the patterns of divergence matrices (KrzCor ranged from 0.74 to 0.90; Table S4). The eigenvector correlations between the two first eigenvectors of pooled P- and D-matrices were very high, ranging from 0.92 to 0.99 (Table S4, Figure S3), whereas correlations were higher only for the first eigenvector when using within-species P-matrices (Table S3). All empirical vector correlations were significant when accounting for uncertainty in estimating the pooled P-matrices and D-matrices, while the same was true for only the first eigenvector when accounting for uncertainty in estimating species P-matrices (Table S4).

The first eigenvector (Egv1), which corresponds to p_{max}, can be interpreted as a size vector: all coefficients had the same sign, indicating that all traits increase or decrease together in this dimension (Jolicoeur 1963). A size vector can be either isometric, in which increases in body size do not change shape, or allometric, in which shape changes with size. To test whether Egv1 was isometric, we divided each of its coefficients by the isometric expectation of 0.408 (i.e., $1/\sqrt{6}$, the expected coefficients for a six-element vector; Jolicoeur 1963). The scaled coefficients for bone lengths were significantly lower than one (except two bones in divergence matrices), whereas those for the square root of muscle CSA, especially from the upper leg, were significantly higher than one (Table S5). This result indicated positive allometry for muscle CSA (stronger within species) and negative allometry for leg length (Figure 2). Therefore, we

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conclude that Egv1 corresponds to allometric size, the relative contribution of each variable to absolute size (Marroig and Cheverud 2005).

Variation along the second eigenvector (Egv2) represented a contrast between bone lengths and muscle CSA (Table S5). This axis can be interpreted as variation in hindlimb shape unassociated with size changes, in which individuals and species with higher muscle CSA at a given body size tended to have relatively shorter legs, compared to individuals and species with lower CSA that tended to have longer legs (for that same body size; Figure 2).

Random drift is insufficient for explaining divergence in hindlimb traits

Regression slopes of the observed morphological divergence on within-species variation were above one for both genera, whether using pooled P-matrices or within-species P-matrices as representing ancestral patterns (except when using *B. boans*; see Table 1). However, only for *Leptodactylus* did the 95% CI of the empirical slopes exclude the 95% CI under BM simulations, which assumed within-species variance-covariance patterns for the rate parameters (Table 1; Figure S4). For *Boana*, the empirical slopes were generally above the 95% CI under BM, but the 95% CI of empirical slopes were broader than *Leptodactylus*. Thus, we could confidently reject random drift as driving species divergence for *Leptodactylus*, but we have less support for rejecting drift in *Boana*. Rejection of drift was mainly driven by higher divergence along Egv1 than expected based on within-species variation in this dimension (Table 1).

Pooled P-matrix and divergence are not aligned with the major axes of the adaptive landscape

The major axes of the AL, represented by the first two eigenvectors of the W-matrix, were not shared between *Boana* and *Leptodactylus* (Table S6). In general, these axes favored opposite

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changes in lengths of specific bones in both genera, and directional net selection has favored opposite changes in bone and muscle CSA only in *Leptodactylus* (Table S6).

Neither the axes of most variation within-species (p_{max} and p_2), nor the axes of most divergence across species (d_{max} and d_2), showed significant vector correlations with the major axes of the AL (Table S7). This result also held for our simplified analysis using just leg length and muscle CSA (Table 2). Whereas both pooled P and divergence matrices showed a significantly high positive correlation between leg length and muscle CSA, the W-matrices showed a very low or negative correlation between these two traits. However, this correlation was only significant in *Leptodactylus* (Table 2).

Evolution of maximum jumping acceleration, but not velocity, tracks allometric size

The model with highest support for maximum acceleration was a BM regression model (Table 3) showing a significant negative relationship between the evolution of maximum acceleration and evolutionary changes in allometric size (d_{max}; Figure 3A). This negative relationship indicated that smaller species, with less proportional muscle CSA, produced higher maximum accelerations than bigger species. An OU regression model with no differences between genera / microhabitat (OU1BM) had modest support for the evolution of maximum acceleration (Table 3), also showing a significant negative evolutionary relationship between maximum acceleration and allometric size.

In contrast, the evolution of jumping peak velocity was unrelated to d_{max} (Table 3, Figure 3B). An intercept-only BM model showed the most support. A two-regime OU model also showed modest support, suggesting a different optimum jumping velocity for arboreal (*Boana*) and terrestrial species (*Leptodactylus*; Table 3).

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Jumping performance gradients are neither aligned with constraints nor with the AL

Very few performance gradients showed significant effects of hindlimb traits on jumping performance when considering all six traits (Table S8). Hindlimb traits explained 22% of the variation in maximum acceleration in *B. boans*, and 16% of the variation in maximum velocity in *L. mystaceus*. In contrast, when analyzing just leg length and muscle CSA, we found more significant relationships (Table S9). Reducing leg length increased maximum acceleration in *B. boans* and across species of *Leptodactylus* (Figure S5). Increasing muscle CSA increased maximum acceleration and velocity in *L. mystaceus*, but we found the opposite pattern across species of *Leptodactylus* (Figure S5). Although not all performance gradients were significant, several of them showed a pattern of opposite signs between leg length and muscle CSA. This result agrees with the low or negative correlation found between these two traits in the W-matrix (Table 2).

The full multivariate performance gradients had very low vector correlations with both p_{max} and p_2 (Table S10). Some performance gradients showed empirical correlations above the null expectation with the first major axis of the AL (*B. boans* for maximum acceleration) or the second major axis of the AL (*L. mystaceus* for maximum velocity). However, the 95% CI were large, making these correlations non-significant (Table S10).

Discussion

We found an overall pattern of hindlimb trait covariation that was highly similar within and between species in both *Boana* and *Leptodactylus* (Table S4), despite a deep divergence of around 60 mya between the genera (Figure 1). Our study adds to the mounting evidence that

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variational patterns within species, reflected by the P- or G-matrix, influence species divergence over long evolutionary timescales (Renaud et al. 2006; Marroig and Cheverud 2005, 2010; Simon et al. 2016; Houle et al. 2017; McGlothlin et al. 2018; Machado 2020, Mongle et al. 2022; Opedal et al. 2023; Machado et al. 2023). However, understanding which factors have shaped both the major axes of the adaptive landscape and those of the G-matrix is challenging. By including functional performance in our study, we were able to infer that the realized adaptive landscape (AL) was not aligned with hindlimb trait correlations, and directional net-selection seemed unaligned with selection associated with jumping performance. Yet, our inference that the adaptive peak of jumping performance randomly drifted across species, coupled with stability of the G-matrix, may explain the persistent effect of patterns of within-species hindlimb trait covariation on divergence.

Stability of genetic architecture of hindlimb traits in both frog genera

We found high similarity among all P-matrices across both genera (Tables S2, S3). This consistency suggests that the G-matrices of all these frog species have remained generally stable over long evolutionary timescales, at least in terms of their major axes of variation. This overall stability of trait covariation seems to be common in nature and has been found in the morphology of many species that have diverged for over tens of millions of years (e.g., Marroig and Cheverud 2001; Porto et al. 2009; Kolbe et al. 2011; Simon et al. 2016; Machado et al. 2018; McGlothlin et al. 2018, 2022).

Such stability of trait correlations may be maintained by selective, functional, or developmental constraints (Arnold 1992). Because we did not find alignment between within-species lines of least resistance and the AL (Table S8), stability of hindlimb trait correlations

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could have been maintained by a stable pattern of correlational selection (Riedl 1977; Cheverud 1984; Brodie 1992; Arnold 2005; McGlothlin et al. 2005; Roff and Fairbairn 2012)—promoting functional integration of trait complexes (Olson and Miller 1958; Hohenlohe and Arnold 2008; Simon et al. 2019)—instead of persistent directional selection acting on mutational effects (Jones et al. 2007, 2014; Pavlicev et al. 2011). We may thus expect correlational selection on hindlimb traits in frogs because unbalanced trait values (e.g., greatly differing leg bone lengths) may hinder locomotion in general, including jumping. Therefore, the positive correlation between leg length and leg-muscle CSA could reflect their coordinated effect on locomotor performance favored by correlational selection. We do emphasize, however, that our estimates of the AL only take net directional selection into account and do not estimate the curvature of the AL (i.e., the W-matrix does not take stabilizing and correlational selection into account). Therefore, the misalignment between the AL and trait correlations does not rule out a role of correlational selection promoting stability of hindlimb trait correlations over evolutionary time.

However, because we found allometric size to be the major axis of variation, the stability of trait correlations likely also reflects, at least partly, developmental constraints. Variation in limb growth rates is regulated by several growth factors related to chondrogenesis and bone formation (Hentschel et al. 2004), and also to chondrocyte proliferation and differentiation (Wang et al. 2004; Cooper et al. 2013). The expression of growth factors is further regulated by several genes that can be expressed with different patterns of timing and spatial positioning (Towers and Tickle 2009). Hence, genetic and developmental variation regulating overall limb growth seems abundant and highly polygenic, and this variation may explain why individuals vary most in allometric size. Allometric size often acts as a line of least evolutionary resistance in several groups, along which species predominantly diverge (Schluter 1996; Marroig and

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Cheverud 2005, 2010; Simon et al. 2016; Machado 2020; Feiner et al. 2021; Tejero-Cicuéndez et al. 2023). This may be due to the ecological relevance of size, in diet for example (Moen and Wiens 2009; Marroig and Cheverud 2010; Machado 2020), but also to conserved developmental regulation of growth patterns, creating developmental lines of least resistance (Salazar-Ciudad 2006). Although developmental lines of least resistance can change over time (Feiner et al. 2021), they likely evolve slowly (Voje et al. 2014; Houle et al. 2019), which fits our findings.

Divergence in allometric size and selection on jumping performance

Our finding that both the G-matrix and divergence were unaligned with the AL (Tables 2, S7) indicates that the high interspecific divergence in allometric size likely reflects genetic and developmental constraints biasing the response to net selection, instead of diversifying selection on allometric size. This interpretation favors the hypothesis that constraint has shaped hindlimb divergence in both genera. In contrast, direct selection on allometric size may not necessarily have shaped hindlimb divergence, although indirect selection on allometric size likely happened due to positive correlations within the hindlimbs.

Our interpretation that selection has influenced hindlimb divergence in *Boana* may be surprising, given the lower support for rejecting drift in hindlimb divergence in *Boana* (Table 1). Our weakest support resulted from using the *B. boans* P-matrix to represent within-species patterns. Yet part of this result may be explained by having fewer individuals of *B. boans*, which can result in overestimation of the first eigenvalue (Hayden and Twede 2002; Meyer and Kirkpatrick 2008). Consequently, higher intraspecific variance on the first eigenvector of *B. boans* P-matrix likely made it harder to reject drift. Consistent with this explanation, the empirical slope for the drift test was higher than the 95% CI for BM simulations when using both

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the pooled P-matrix and *B. raniceps* P-matrix, for which we had higher sampling than in *B. boans*. Moreover, the large 95% CI for the empirical regression slopes likely reflect a low number of traits (and, therefore, eigenvectors) to estimate the regression (six in total). Hence, despite the lower support for rejecting drift in *Boana*, we also infer selection as the main driver of divergence in this genus.

Moreover, our analysis of performance gradients showed further evidence for selection on hindlimb traits (Tables S8, S9, Figure S5). Nonetheless, the performance gradients were generally neither aligned with directional net-selection nor with hindlimb divergence (Tables 2, S10). These results may indicate that variation in jumping performance was not a major contributor to differences in fitness across the species over evolutionary time. While seemingly counterintuitive for organisms that primarily move by jumping, this interpretation is consistent with previous work. For example, frogs showed only a single optimum for jumping performance despite diversity in limb shape at a broader evolutionary scale, suggesting many-to-one mapping of morphology onto performance (Moen 2019). However, we cannot discard poor estimation of the AL (W-matrix) and of within-species performance gradients as contributing factors to this misalignment. For the estimation of the W-matrix, a potential issue is the necessity of inverting the pooled P-matrix. Such a procedure makes the trailing eigenvalues, which are poorly estimated, dominate the inverted matrix (Hayden and Twede 2002). We tried to account for this issue by using an eigenvalue extension procedure (replacing the sixth eigenvalue of the pooled P-matrix by the fifth, which is better estimated) on the pooled P-matrix, following Marroig et al. (2012), to better estimate the W-matrices (although this procedure is recommended for matrices with 10 or more traits). Yet, the major axes of the AL were still unaligned with within-species variation and with performance gradients (results shown in ‘R script 1.R’; Simon et al. 2025).

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Moreover, other unmeasured traits may be important for jumping, potentially obscuring the relationships between performance gradients and the AL.

Although our evolutionary analyses of maximum jumping velocity and acceleration across species in *Boana* and *Leptodactylus* suggest that jumping performance evolved randomly (Table 3), this pattern can alternatively be interpreted as jumping performance evolving by directional selection to track an adaptive peak that itself has moved randomly (Felsenstein 1988; Hansen and Martins 1996; Arnold et al. 2001; Hansen 2012; Holstad et al. 2024; Uyeda and McGlothlin 2024). We favor this adaptive explanation for the evolution of jumping performance because of its relevance in frog ecology (Gans and Parsons 1966; Emerson 1978; Duellman and Trueb 1994; Marsh 1994; Moen 2019). Adaptive peaks may move in a random fashion when the environment fluctuates, and these fluctuations in the direction of selection may explain a persistent effect of genetic variance on divergence if species mostly track the peaks along genetic lines of least resistance (Holstad et al. 2024). Such fluctuations in the jumping performance peak may relate to changes in the balance of functional trade-offs (Garland et al. 2022) on the hindlimbs, which also influence other locomotor behaviors (e.g., swimming, climbing, burrowing, walking; Moen et al. 2013; Reynaga et al. 2018; Moen 2019; Mendoza et al. 2020).

Evolution of maximum jumping acceleration and allometric size

The negative relationship between the evolution of maximum acceleration and allometric size suggests a functional constraint associated with size scaling. Generally, larger animals have larger muscles, which produce higher forces. These forces depend on muscle CSA, which scales to the second power of length. However, such larger animals have proportionally larger masses to move with those muscles, because mass scales to the third power of length. Overall, this

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scaling could result in lower whole-body acceleration in larger animals. One potential consequence is the positive allometry for muscle CSA that we found within and across species (Table S5). If larger animals have proportionally more muscle CSA for their size, they may be able to partially offset the reduced jumping acceleration due to muscle CSA-body mass scaling.

However, the pattern of smaller species showing higher peak jumping acceleration than larger species may also be interpreted as an evolutionary innovation in response to size evolution in frogs. Smaller species may more frequently use power amplification by elastic energy storage to achieve higher accelerations, despite having lower absolute muscle mass, than larger species (Astley 2016; Ilton et al. 2018; Sutton et al. 2019; Mendoza et al. 2020). Those smaller animals, with shorter legs and thus contact times, need higher acceleration than larger animals to achieve a comparable take-off velocity. Yet such higher acceleration is limited by both their reduced muscle CSA and the reduced force produced by fast muscular contractions (which are necessary if the contact time is very short in small animals; Vogel 2013). On the other hand, such small species may be more likely to release elastic energy stored in hindlimb connective tissue (Sutton et al. 2019) and do so more effectively (Mendoza and Azizi 2021). Hence, species divergence being restricted mostly to allometric size may have strongly affected the evolution of jumping acceleration, as an evolutionary solution in smaller species to a functional constraint of muscle function and leg length. This evolutionary innovation may also explain why the performance gradients in *L. mystaceus* showed an opposite pattern than within the genus *Leptodactylus* (Figure S5). While bigger muscles produce more force to increase acceleration and velocity within species (Vogel 2013, Biewener and Patek 2018), this is not the pattern seen across species when smaller species may be more likely and more effective in releasing elastic energy to

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achieve higher locomotor accelerations and velocity (Sutton et al. 2019; Mendoza and Azizi 2021).

Conclusions

We found support for the constraint hypothesis in shaping long-term divergence in hindlimb traits on frogs, with a potentially strong effect of conserved developmental constraints, channeling divergence mostly to allometric size. This pattern is consistent with development having a primary role in aligning micro- and macroevolution, especially in a view of a dynamic adaptive landscape. In this view, fluctuating selection on functional performance can reinforce a link between these scales over larger timescales by making species phenotypic means chase these adaptive peak fluctuations mostly along genetic lines of least resistance. However, developmental constraints can be broken by episodes of strong selection, which may occur when lineages invade a new environment that demands great changes in specific behaviors, such as locomotion (i.e., a new adaptive zone). Therefore, studying anuran species that differ more strongly in ecology and performance, such as aquatic and burrowing species, may favor a pure-selection hypothesis and a role of selection on performance in shaping both genetic architecture of traits and species' phenotypic means. Overall, it is likely that both development and function play a role in long-term phenotypic divergence. However, knowing how often each of these factors dominate macroevolutionary patterns demands directly studying the macroevolutionary adaptive landscape and its connection with both developmental and functional processes acting within-populations and species.

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Acknowledgements

We thank Anne-Claire Fabre for help in collecting frogs in 2015, Menelia Vasilopoulou-Kampitsi for help in early analysis of force traces, and Craig McGowan for assistance in smoothing force traces. Antoine Fouquet contributed a key specimen for performance testing and helped with identification of specimens and systematics of anurans from French Guiana. Finally, for loans of museum specimens, we thank Jeremy Jacobs, Kevin de Queiroz, and Rob Wilson at the Division of Amphibians and Reptiles at the Smithsonian Institution. We also thank editor-in-chief Volker Rudolf and associate editor Joel McGlothlin for great suggestions on our manuscript, and two anonymous reviewers that helped improve our work.

Statement of Authorship

Conceptualization: D.S.M., A.H., M.N.S.; Funding acquisition: D.S.M, A.H, E.A.C..; Experimental design: D.S.M., A.H.; Data collection: E.A.C., D.S.M, A.H., M.N.S.; Data analysis: M.N.S., D.S.M.; Coding: M.N.S.; Data validation and visualization: M.N.S.; Writing – original draft: M.N.S., D.S.M.; Writing – review and editing: M.N.S., D.S.M., A.H., E.A.C.

Data and Code Accessibility

The data associated with this work is deposited at Dryad (Simon et al. 2025; https://doi.org/10.5061/dryad.rn8pk0pnp) and the code is deposited at Zenodo (https://doi.org/10.5281/zenodo.14757462; also linked with the Dryad submission).

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Tables

Table 1. Random drift tests for hindlimb traits in the two frog genera.

Drift test	empirical slope	empirical 95% CI	BM 2.5%	simulations 97.5%
Pooled P				
Boana	*1.35*	0.77, 1.92	0.87	1.13
Leptodactylus	**1.44**	1.14, 1.73	0.85	1.14
Within-species P				
Boana (*B. boans*)	1.00	0.69, 1.32	0.90	1.10
Boana (*B. raniceps*)	*1.30*	0.75, 1.85	0.85	1.15
Leptodactylus	*1.41*	0.89, 1.93	0.86	1.15
Drift test excluding p_{max}				
Pooled P				
Boana	0.94	0.21, 1.67	0.78	1.21
Leptodactylus	*1.25*	0.64, 1.86	0.71	1.24
Within-species P				
Boana (*B. boans*)	0.72	0.48, 0.96	0.83	1.18
Boana (*B. raniceps*)	1.03	0.05, 2.01	0.76	1.23
Leptodactylus	*1.21*	0.00, 2.43	0.72	1.15

Note: We performed a regression test of within-species variation on between-species divergence. We strongly rejected drift (values in bold) when the empirical slope 95% CI was outside of the 95% CI of Brownian-motion (BM) simulations that used the corresponding P-matrices as the rate matrix. Values in italic indicate empirical slopes that were higher than the 95% CI of BM simulations. We also performed the test when excluding variation in the first eigenvector (p_{max}) to show that drift was rejected in the overall (two-axis) drift test only because of higher-than-expected variation along p_{max}, rather than both axes.

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Table 2. Pooled P-, divergence, and W-matrices when using total leg length and muscle CSA.

Pooled P	Pooled *Boana*		Pooled *Leptodactylus*	
	leg length	muscle CSA	leg length	muscle CSA
leg length	**0.004**	**0.660**	**0.008**	**0.758**
	(0.0002, 0.006)	(0.602, 0.788)	(0.005, 0.0086)	(0.626, 0.830)
muscle CSA	**0.004**	**0.010**	**0.010**	**0.021**
	(0.0026, 0.0065)	(0.006,0.012)	(0.0055, 0.010)	(0.013, 0.021)
Divergence	*Boana*		*Leptodactylus*	
	leg length	muscle CSA	leg length	muscle CSA
leg length	**0.004**	**0.939**	**0.005**	**0.957**
	(0.0023, 0.0059)	(0.884, 0.973)	(0.0028, 0.0086)	(0.911, 0.980)
muscle CSA	**0.004**	**0.006**	**0.006**	**0.008**
	(0.0024, 0.0069)	(0.0033,0.0089)	(0.0033, 0.010)	(0.0044, 0.014)
W-matrix	*Boana*		*Leptodactylus*	
	leg length	muscle CSA	leg length	muscle CSA
leg length	**3.565**	0.025	**5.685**	**-0.383**
	(2.06, 4.40)	(-0.18, 0.50)	(3.83, 9.13)	(-0.72, -0.13)
muscle CSA	0.036	**0.557**	**-0.677**	**0.550**
	(-0.32, 0.52)	(0.33, 0.92)	(-2.40, -0.17)	(0.32, 1.45)

Note: Pooled P-matrices show within-species trait patterns, whereas divergence matrices show between-species patterns calculated using phylogenetic independent contrasts (i.e., standardized to time). W-matrices show patterns of co-selection between the traits. Diagonal elements are variances, below-diagonal elements are covariances, and above-diagonal elements are correlations. Values in parentheses correspond 95% CI from bootstrapped matrices. Bold values are significantly higher or lower than zero under $\alpha = 0.05$.

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Table 3. Model comparison of BM and OU models for maximum jumping acceleration and velocity.

Univariate models				
Max acceleration	k	loglik	AICc	AICc weight
BM	2	-3.79	12.40	0.02
OU1	3	-3.35	14.40	0.01
OU2	4	-3.17	17.40	0.00
BM regression[a]	**3**	**1.33**	**5.05**	**0.69**
OU1BM[b]	**4**	**1.99**	**7.10**	**0.25**
OU2BM	5	1.99	11.00	0.03
Max velocity				
BM[c]	**2**	**6.08**	**-7.35**	**0.51**
OU1	3	6.10	-4.48	0.12
OU2[d]	**4**	**8.29**	**-5.50**	**0.20**
BM regression	3	6.08	-4.44	0.12
OU1BM	4	6.10	-1.12	0.02
OU2BM	5	8.29	-1.57	0.03

Note: Intercept-only models, BM and OU1, have no predictors. Regression models have species' scores on the axis of most divergence—allometric size—as the predictor. The OU regression models can be single (OU1BM) or two-optima models (OU2BM: arboreal or terrestrial), which correspond to one or two intercepts, respectively. Models in bold show AICc weights over 0.2. k = number of parameters; loglik = log-likelihood of the model; AICc = Akaike Information Criterium for small sample sizes; AICc weight = relative support for each model.

Parameter estimates of most supported models (MLE with 95% CI): [a] σ^2 = 0.046 (0.011, 0.15), slope = -0.21 (-0.32, -0.10); [b] half-life = 21% of the tree length (0%, 199%), slope = -0.22 (-0.32, -0.10); [c] σ^2 = 0.037 (0.016, 0.097); [d] half-life = 14% of the tree length (0.001%, 500%); optima: arboreal = 1.10 m/s (1.00, 1.21); terrestrial = 0.83 m/s (0.71, 0.93).

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Figure legends

Figure 1. Frog species in the genera *Boana* and *Leptodactylus* for which we collected only morphological data (60 species) or both morphological and jumping performance data (18 species, in bold). Morphological data were bone lengths (shown here as their sum, leg length) and muscle CSA of hindlimbs. Performance data were maximum jumping acceleration and velocity. Boxplots show the median (lines), interquartile range (boxes), 95% distribution (whiskers), and points outside the 95% distribution.

Figure 2. Species distributions along axes of most phenotypic divergence for *Boana* (A) and *Leptodactylus* (B). The first axis corresponds to d_{max} and represents variation in allometric size, which explains 95% and 97% of divergence across species in *Boana* and *Leptodactylus*, respectively. The second axis corresponds to d_2 and represents hindlimb shape, as a contrast between bone length and muscle CSA. d_2 explains only 2% and 3% of divergence across species in *Boana* and *Leptodactylus*, respectively. These divergence axes are similar to axes of most variation within species. Arrows point from dots of species in photos, which are not scaled to body size. All photos by DSM.

Figure 3. Scatterplots of maximum jumping acceleration (A) and velocity (B) versus allometric size. We ran different evolutionary models in *slouch*, assuming a BM or an OU model for jumping performance. While the BM regression was the best supported model for maximum acceleration (Table 3; σ^2 [MLE ± SE] = 0.046 ± 0.010, slope = -0.21 ± 0.05, R^2 = 0.48), an intercept-only BM model showed the highest support for maximum velocity (Table 3; σ^2 = 0.037 ± 0.010).

Figure 1

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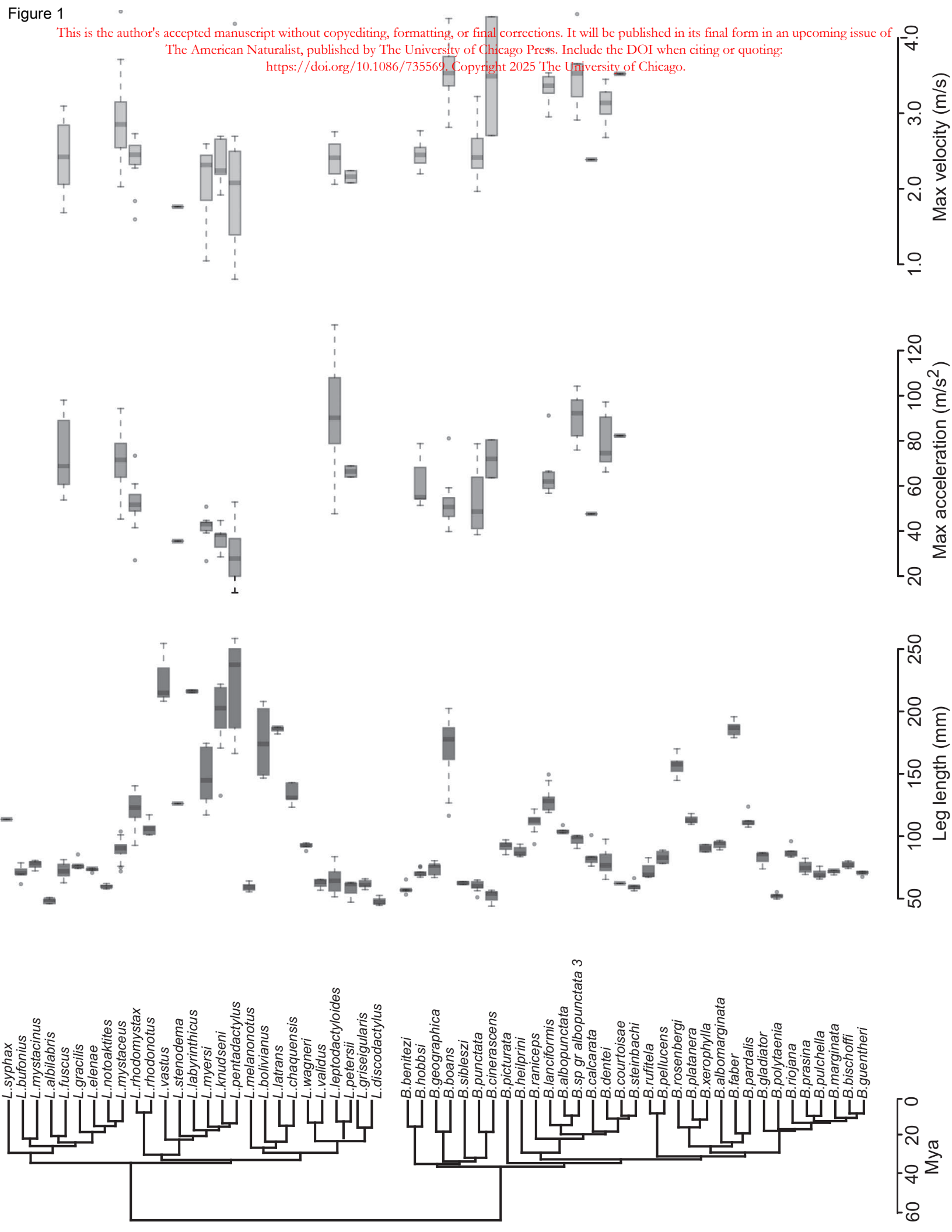

Figure 2 (print only)

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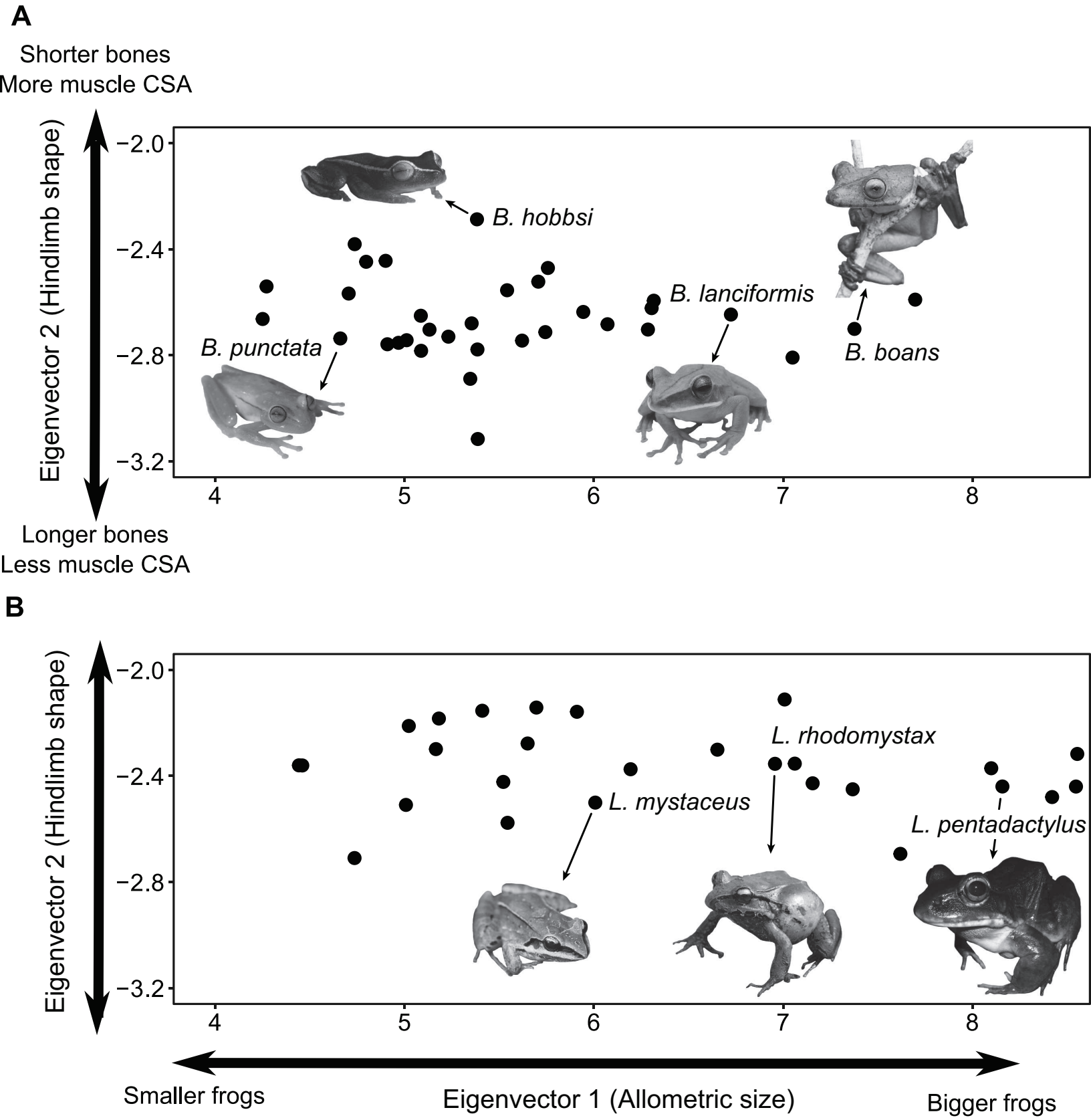

Figure 2

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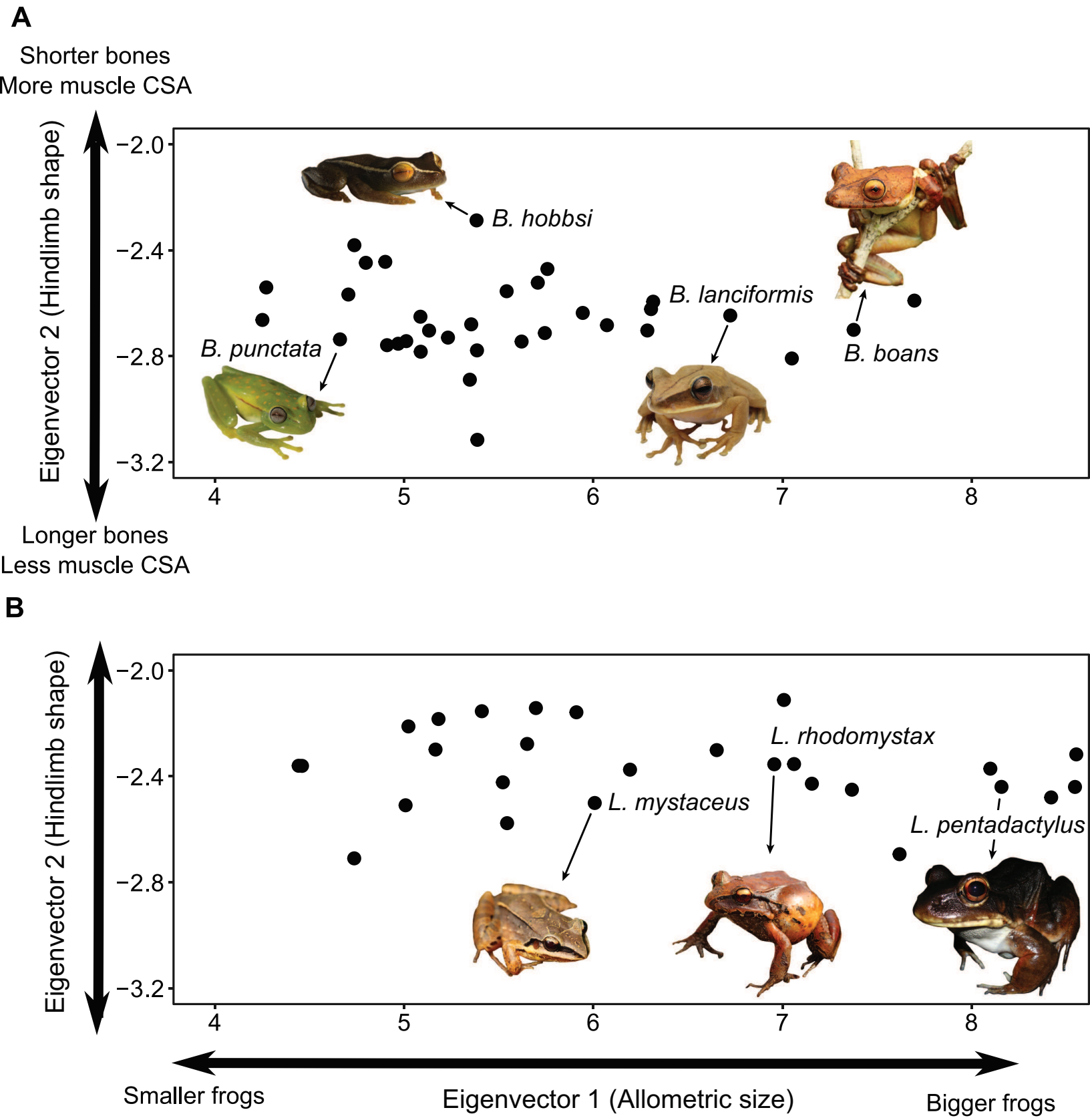

Figure 3

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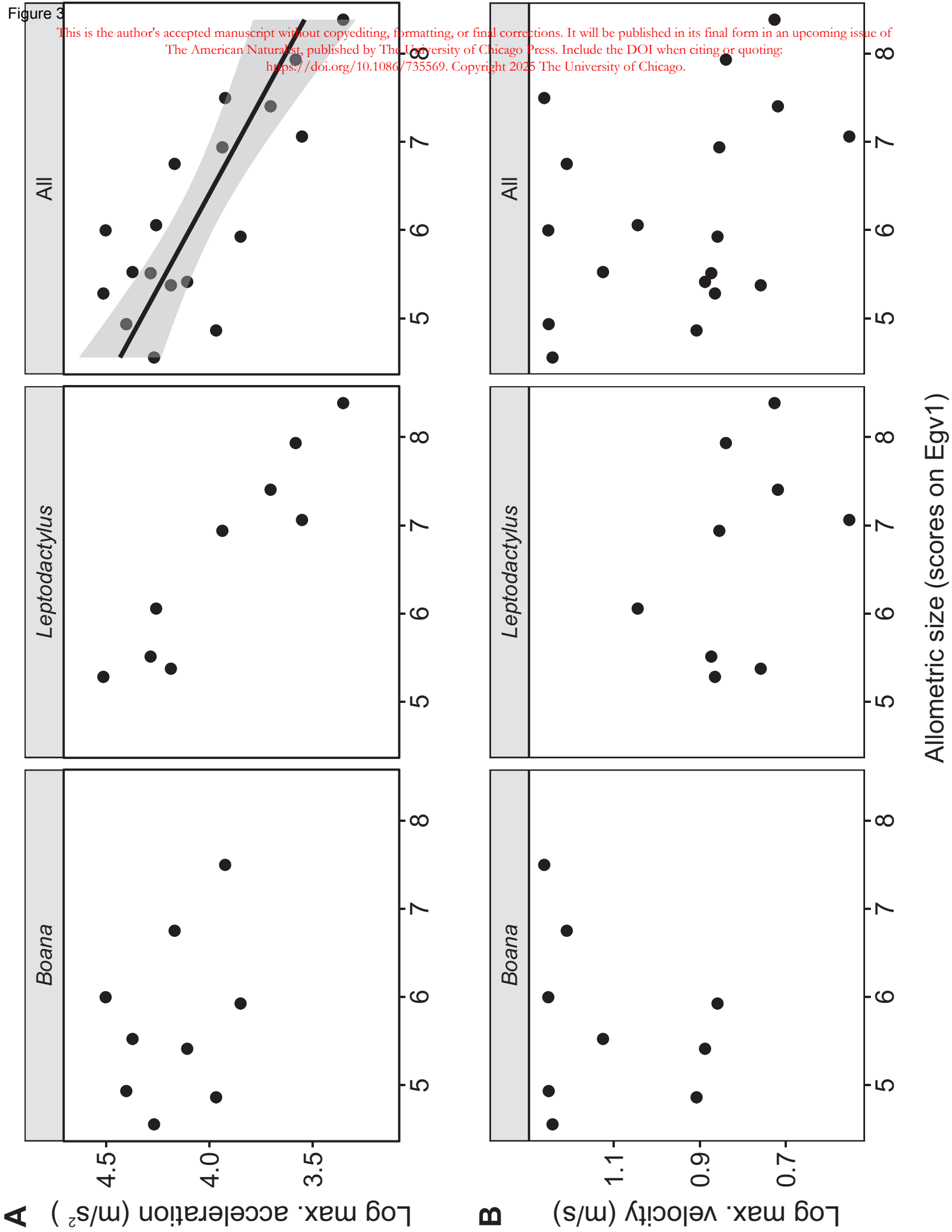

Supplementary Files (PDF, Word, TeX, figures)

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Online Supplement

Macroevolutionary divergence along allometric lines of least resistance in frog hindlimb traits and its effect on locomotor evolution

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Journal: American Naturalist

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Supplementary Methods, Tables and Figures

Additional detail for analysis of force traces

We used a MATLAB script (version R2020a, MathWorks) to smooth the X, Y, and Z force traces independently, after extracting them from the BioWare software (version 5.4.8.0, Kistler). We applied a moving-average filter and manually adjusted filtering weights to reduce noise without flattening the (real) peaks (Schwaner et al. 2021). After smoothing, we centered the force traces to zero prior to movement and calculated acceleration by dividing traces by body mass in kilograms. We then obtained velocity-by-time profiles by integrating the acceleration profiles, using cumulative trapezoidal numeric integration. We finally used the Pythagorean theorem to calculate resultant profiles in MATLAB (i.e., single-dimensional velocity and acceleration profiles).

For some force traces, the start of the peaks (needed for proper integration) was unclear due to perturbations just before the jump, seen as ‘bumps’ in the traces. We could not remove these perturbations by smoothing the curves because additional smoothing excessively flattened the acceleration peaks. Nonetheless, these perturbations produced excessively long contact times (above 0.15 s) for some curves, resulting in unrealistically high velocities (above 4.5 m/s, the highest published take-off velocity in any frog; James and Wilson, 2008).

To address this issue, we adopted a cut-off procedure to decide when contact times were unrealistically high for a given individual. We compared contact times of all acceleration curves for an individual, and compared each with an individual's mean contact time. We adopted different cut-offs to determine when a contact time was too long: 20%, 40%, or 50% higher than mean contact time. Using the least stringent cut-off, we only eliminated four acceleration curves

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from a total of 427 curves, and were still left with unrealistically high velocity values. The most stringent cut-off resulted in elimination of 13 curves. Using this cut-off (50% higher than mean) eliminated most of the unrealistically high maximum velocities, but we were still left with eight values above 4.5 m/s, which is still almost 50% higher than the range previously found for these genera (Moen et al. 2013). To be conservative (i.e., not include data that were in error), we excluded these force traces with abnormally high velocity values (above 4.5 m/s). These excluded traces summed to only 23 profiles out of a total of 427. Figures S1 and S2 show the final distribution of contact times and maximum velocities we used for further analysis.

Matrix similarity methods: Random Skewers and Krzanowski Projection

We used two methods to compare intraspecific P-matrices and pooled P-matrices: Random Skewers (RS) and the Krzanowski correlation (Krz). RS uses the multivariate response to selection equation ($\mathbf{\Delta z} = \mathbf{G\beta}$; Lande 1979) to measure the vector correlation of 1,000 response vectors ($\mathbf{\Delta z}$) between pairs of P-matrices after both are multiplied by 1,000 unit-length random selection vectors (i.e., β vectors whose sum of squared elements equals 1; Cheverud and Marroig, 2007). A high correlation indicates high similarity in the response to selection between P-matrices, which is determined by patterns of trait covariation and their associated variances. Significance is tested against a null distribution of vector correlations using random vectors drawn from a multivariate normal distribution with a mean of zero and standard deviation of 1. The upper 5% tail of a distribution of random six-element vector correlations was 0.76.

The Krz method measures matrix similarity by correlating the best-matched pairs of eigenvectors of a pair of P-matrices (Krzanowski 1979). The method calculates a projection matrix, based on the subspace of a set of eigenvectors of each matrix that represents the minimal

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angles between these eigenvectors. In general, half the number of eigenvectors minus 1 is used in this method (Krzanowski 1979). In our case, this meant two best-matched pairs of eigenvectors for morphological P-matrices. To be conservative, we chose to use the best-matched three pairs of eigenvectors to compare intraspecific P-matrices, and four pairs to compare P-matrices and divergence matrices. The sum of the eigenvalues of this projection matrix can be used to determine the similarity of the eigenvector subspaces between the pair of matrices being compared (Blows et al. 2004). In our case, we divided the sum of the eigenvalues by the number of eigenvectors used (four), such that the sum of the eigenvalues may equal 0 (no similarity) or 1 (full similarity). A high Krz correlation indicates shared patterns of trait covariation. However, unlike RS, it does not account for how much variance the eigenvectors explain in the traits. Results are shown in Table S2.

Taxon substitutions in the phylogeny

Most of our sampled taxon were in the phylogeny of Portik et al. (2023). However, four species that we sampled for morphology and/or performance were not included. Therefore, we changed the tip labels for four substitute taxa in the phylogeny of Portik et al. (2023) in order to analyze the data from our sampled taxa. We expect these substitutions to minimally affect our phylogenetic comparative analyses, because the substitutions were either arbitrary (e.g., substituting a sister taxon; cases (i) below) or only slightly changed the expected branch lengths (e.g., cases (ii) and (iii) below).

Nonetheless, we describe these substitutions here and their potential effects on our analyses: (i) *Boana crepitans* (phylogeny) for *B. platanera* (phenotypic data), given that the latter was recently split from the former (Orrico et al. 2017; Escalona et al. 2021) and is the sister

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taxon to *B. xerophylla*; (ii) *Boana multifasciata* (phylogeny) for *B. sp.gr. albopunctata* 3 (phenotypic data; see Fouquet et al. 2021), given that the latter is from an undescribed species complex (Fouquet et al. 2021) for which *B. multifasciata* is the best substitute in Portik et al. (2023); and (iii) *Boana alfaroi* (phylogeny) for *B. courtoisae* (phenotypic data), since *B. alfaroi* and *B. tetete* are the most closely related taxa sampled by Portik et al. (2023) to *B. courtoisae* (Fouquet et al. 2021), given we also have morphological data for *B. steinbachi*. Note that while specimens now assigned to *B. courtoisae* were formerly *B. fasciata*, specimen sampling locations suggest that the genetic data for the latter species in Portik et al. (2023) are a chimera of at least two species, including *B. courtoisae* and *B. steinbachi* (Fouquet et al. 2021), making it an unsuitable substitute. Instead, choosing *B. alfaroi* or *B. tetete* as the substitute for *B. courtoisae* produced relationships best aligned with those of Fouquet et al. (2021), who analyzed much finer-grained geographical sampling within the *Boana albopunctata* group. Finally, we confirmed or updated species names based on collection localities and recent taxonomic literature (Koehler et al. 2010; de Sá et al. 2014; Orrico et al. 2017; Schneider et al. 2019; Caminer and Ron 2020; Da Silva et al. 2020; Magalhães et al. 2020; de Almeida et al. 2021; Faivovich et al. 2021; Carvalho et al. 2022; Marinho et al. 2022)

Alternative evolutionary models to calculate the divergence matrix

We acknowledge that assuming a BM model of evolution may not be the optimal evolutionary model to characterize the evolution of hindlimb traits and jumping performance in frogs, which are expected to be under long-term selection (Moen 2019). Hence, we also estimated D-matrices after rescaling the phylogeny assuming a single-peak Ornstein-Uhlenbeck model of evolution, which models adaptive evolution as a stochastic process with a deterministic change in species'

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means towards a central value (Felsenstein, 1988; Hansen and Martins, 1996). D-matrices assuming BM and OU were highly similar for both genera (morphology: average RS = 0.99, average Krz = 0.99 for both genera; morphology + performance: average RS = 0.99, average Krz = 0.99 for *Leptodactylus*; average RS = 0.98, average Krz = 0.95 for *Boana*). Therefore, we only use D-matrices calculated using the mean squares and cross-products of PICs (i.e., under BM) for further analyses, because it is simpler, with fewer parameters.

Bootstrapping procedures

We used bootstrapping to calculate several of the 95% CI of vector correlations between eigenvectors of pooled P-matrices and divergence matrices; between phenotypic lines of least resistance (p_{max} and p_2) with axes of the realized adaptive landscape (w_{max} and w_2); and between performance gradients with p_{max}, p_2, w_{max}, and w_2. For pooled P-matrices, we bootstrapped individuals, with the same sample size as original matrices, to create 1,000 resampled data sets. We then used linear models of hindlimb traits on sex and species to re-estimate 1,000 pooled P-matrices for each frog genus. For divergence matrices, we performed parametric bootstrapping 1,000 times using the function "sim.char" from the R package *geiger* (Harmon et al. 2008), assuming a BM model, the empirical interspecific variation as the rate matrix, and the species phylogeny. We then re-estimated the rate matrix from the simulated data sets of species average values of hindlimb traits, using the function "ratematrix." Finally, to obtain resampled distributions of the W-matrices, we calculated them with equation 2 and used the resampled pooled P-matrices and simulated divergence matrices.

Random drift test

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To perform the drift tests, we regressed variation across species (projected on eigenvectors of the pooled P-matrix) on eigenvalues of the pooled P-matrix (which represented the within-species matrix), following Ackermann and Cheverud (2002):

$$\ln D_i = \ln(t/N_e) + \beta \ln(P_i)$$

Here, P_i is the within-population variance of its *i*th eigenvector (i.e., its eigenvalue) and D_i is the variance of species' means projected on the eigenvectors of the P-matrix (i.e., the variance of species scores on P-matrix eigenvectors). For example, D_1 is the amount of observed species divergence in the direction of most within-species variation (i.e., P_1).

If the slope is lower than one, it indicates less divergence than expected by drift, suggesting stabilizing selection. If the slope is higher than one, it indicates higher divergence than expected by drift, suggesting diversifying selection (Ackermann and Cheverud, 2002; Marroig and Cheverud, 2004). We performed the drift test using the eigenvalues of the pooled P-matrix of *Boana* or *Leptodactylus*, but also the eigenvalues of species-specific P-matrices, and the PICs of species' means projected onto the pooled or species-specific P-matrix eigenvectors. Using PICs accounted for the phylogenetic non-independence of species means. Moreover, using well-estimated pooled P-matrices as the ancestral matrix (rather than estimated from a single species) has the advantage of lower or no bias in the first eigenvalue, which is generally overestimated when sample sizes are small (Opedal et al. 2023). We also calculated the 95% CI for the empirical regression slope (using the function 'confint' on the regression model).

Instead of relying on the parametric confidence interval of β to test drift, we simulated species' mean trait values 1,000 times on each genus-level phylogeny assuming a BM model of evolution, given that it can be interpreted as a random drift process (Hansen and Martins 1996). We used the pooled P-matrix of *Boana* or of *Leptodacylus* for the rate matrix in the function

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"sim.char" from the R package *geiger*, so that species divergence would arise from the patterns of variation and covariation we assumed as those of the ancestral population in the drift test. With the 1,000 simulated data sets, we calculated the PICs of each species and repeated the drift test using the eigenvalues of the pooled P-matrix and the variance of species' scores projected on each eigenvector of that same matrix. We then calculated the 95% CI of β under drift using the 1,000 slopes obtained from the simulations (Simon et al. 2016). We rejected drift when the 95% CI of the empirical regression slope fell outside the simulated 95% CI, which is the range expected for the slope when assuming drift as the underlying process of species divergence. Given that this test of drift uses the eigenvalues for the regression, the 95% CI of the empirical regression slope may be broad, as in our case of six traits (and thus six eigenvalues for estimating the regression slope). Thus, we emphasize that our criterion for rejecting drift produces a very conservative test. Figure S4 shows the regression for each genus when using the pooled P-matrix or species-specific P-matrices as the within-species patterns.

Multivariate analysis of selection on performance

Both maximum jumping velocity and acceleration were correlated when joining the 18 species (across genera) with these data ($r = 0.60$; using "phyl.vcv" from the package *phytools*; version 2.1.1, Revell 2012). Thus, we first performed a multivariate analysis with both performance variables as response variables and species' scores on d_{max} as the predictor, using the package *mvSLOUCH* (version 2.7.6, Bartoszek et al., 2012, 2023). However, the simplest multivariate OUBM models possible could not be well estimated with our species sample size (e.g., negative adjusted R-squared values), precluding a proper model comparison. Thus, we chose to perform univariate analysis of selection on performance, as described in the main text.

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Performance gradients

The relationship between within-species gradients and selection within populations is more straightforward to interpret than across species. Yet the comparison of the major axes of the AL with between-species gradients may be more relevant to indicate whether shifts in morphological adaptive peaks relate to species' differences in jumping. Thus, for between-species performance gradients, we fit phylogenetic multiple linear regressions within *Boana* or *Leptodactylus*, assuming BM (for residual error) and adding intraspecific standard errors to the jumping variables. We ran interspecific analyses with the function "phylolm" from the *phylolm* R package (version 2.6.2; Ho and Ané 2014). We also ran analyses that calculated the maximum-likelihood value of lambda for residual error and obtained similar gradients.

For relationships between species, we only found that reducing femur length increased maximum velocity in *Leptodactylus* (Table S8). In contrast, when analyzing just leg length and muscle CSA, we found a higher number of significant relationships (Table S9). Reducing leg length increased maximum acceleration in *B. boans* and across species of *Leptodactylus* (Figure S5). Increasing muscle CSA increased maximum acceleration and velocity in *L. mystaceus*, but we found the opposite pattern across species of *Leptodactylus* (Figure S5).

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Table S1. Sample sizes for all species studied, separated by sex. We sampled 33 species in *Boana* and 27 in *Leptodactylus* for morphology, and 9 species of each genus for both morphology and jumping performance data (in bold).

Species	F	M	Total	Species	F	M	Total
B. albomarginata	0	5	5	*L. albilabris*	0	5	5
B. albopunctata	0	5	5	*L. bolivianus*	0	4	4
B. benitezi	0	5	5	*L. bufonius*	0	5	5
B. bischoffi	0	5	5	*L. chaquensis*	0	5	5
B. boans	13	32	45	*L. discodactylus*	0	4	4
B. calcarata	1	4	5	*L. elenae*	0	5	5
B. cinerascens	1	2	3	***L. fuscus***	3	10	13
B. courtoisae	0	1	1	*L. gracilis*	0	5	5
B. dentei	2	5	7	*L. griseigularis*	0	5	5
B. faber	0	5	5	***L. knudseni***	2	2	4
B. geographica	0	5	5	*L. labyrinthicus*	0	2	2
B. gladiator	0	3	3	*L. latrans*	0	3	3
B. guentheri	0	5	5	***L. leptodactyloides***	3	15	18
B. heilprini	0	5	5	*L. melanonotus*	0	5	5
B. hobbsi	2	14	16	***L. myersi***	1	4	5
B. lanciformis	2	8	10	***L. mystaceus***	12	40	52
B. marginata	0	3	3	*L. mystacinus*	0	5	5
B. pardalis	0	5	5	*L. notoaktites*	0	5	5
B. pellucens	0	5	5	***L. pentadactylus***	2	9	11
B. picturata	0	5	5	***L. petersii***	1	5	6
B. platanera	0	3	3	***L. rhodomystax***	3	14	17
B. polytaenia	0	5	5	*L. rhodonotus*	0	5	5
B. prasina	0	4	4	***L. stenodema***	1	0	1
B. pulchella	0	5	5	*L. syphax*	0	1	1
B. punctata	6	11	17	*L. validus*	0	5	5
B. raniceps	1	43	44	*L. vastus*	0	3	3
B. riojana	0	6	6	*L. wagneri*	0	5	5
B. rosenbergi	0	5	5				
B. rufitela	0	5	5				
B. sibleszi	0	2	2				
B. sp. gr. albopunctata 3	0	6	6				
B. steinbachi	0	5	5				
B. xerophylla	0	2	2				

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Table S2. P-matrix similarity among within-species and pooled P-matrices under two methods. Both methods use vector correlations to measure similarity. Random Skewers (RS) gives the mean correlation of response vectors when applying the same random selection vectors to a pair of matrices. Significance is given by the empirical value being above the 95% percentile of a null expectation of the correlation for random vectors with the same number of elements (0.76); all values were significant. The Krzanowski correlation gives the sum of the correlations between the three best-matched eigenvectors of a pair of matrices divided by the number of correlations, thus ranging from zero (no similarity) to one (complete similarity).

RS	*B. boans*	*B. raniceps*	*L. mystaceus*	*Boana*
B. raniceps	0.89			
L. mystaceus	0.93	0.94		
Boana	0.97	0.96	0.96	
Leptodactylus	0.95	0.93	0.98	0.97
Krz	*B. boans*	*B. raniceps*	*L. mystaceus*	*Boana*
B. raniceps	0.97			
L. mystaceus	0.97	0.93		
Boana	1.00	0.96	0.97	
Leptodactylus	0.95	0.91	0.97	0.96

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Table S3. First and second eigenvectors of morphological pooled P-matrices and divergence matrices. The pooled P-matrices were estimated using all individuals from each genus and removing effects of sex and species on trait means. Divergence matrices were represented as rate matrices using a Brownian Motion model of evolution and species morphological means. Numbers are the eigenvector coefficients, which indicate the contribution of each trait to that dimension. "% variance" is the percentage variance of the data explained by each eigenvector.

Pooled P-matrices	Pooled *Boana*		Pooled *Leptodactylus*	
	Egv1	Egv2	Egv1	Egv2
femur	0.326	-0.433	0.355	-0.443
tibia	0.331	-0.356	0.328	-0.357
tarsus	0.330	-0.363	0.328	-0.402
foot	0.371	-0.306	0.300	-0.267
up. leg muscle CSA	0.550	0.445	0.536	0.574
low. leg muscle CSA	0.484	0.512	0.531	0.337
% variance	76	13	83	11
Divergence matrices	*Boana*		*Leptodactylus*	
	Egv1	Egv2	Egv1	Egv2
femur	0.372	-0.224	0.385	-0.307
tibia	0.377	-0.408	0.378	-0.340
tarsus	0.339	-0.530	0.374	-0.263
foot	0.402	-0.117	0.357	-0.398
up. leg muscle CSA	0.467	0.534	0.465	0.399
low. leg muscle CSA	0.475	0.452	0.475	0.635
% variance	95	3	97	2

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Table S4. Similarity and eigenvector correlations between pooled P-matrices or within-species P-matrices and divergence matrices. Table shows KrzCor similarity using four pairs of eigenvectors and empirical vector correlations between eigenvector 1 (p_{max}) of pooled P-matrices and the first axis of highest variation in divergence matrices (d_{max}), and between eigenvector 2 (p_2) of pooled P-matrices and the second axis of most variation in divergence matrices (d_2). Significance was tested using the 95% CI for the vector correlations, estimated by bootstrapping. The correlation was considered significant (in bold) when the 95% CI excluded the expected value of 6-element vector correlations by chance (0.76).

Matrix similarity between P and D		**Vector correlations**			
Pooled P		p_{max} x d_{max}	empirical	CI 2.5%	CI 97.5%
Boana	0.88	Pooled P			
Leptodactylus	0.74	*Boana*	**0.99**	0.98	1.00
Within-species P		*Leptodactylus*	**0.99**	0.98	1.00
Boana (*B. boans*)	0.90	Within-species P			
Boana (*B. raniceps*)	0.83	*Boana* (*B. boans*)	**0.99**	0.97	1.00
Leptodactylus	0.77	*Boana* (*B. raniceps*)	**0.96**	0.86	0.98
		Leptodactylus	**0.98**	0.95	0.99
		p_2 x d_2	empirical	CI 2.5%	CI 97.5%
		Pooled P			
		Boana	**0.94**	0.84	0.97
		Leptodactylus	**0.92**	0.83	0.96
		Within-species P			
		Boana (*B. boans*)	0.82	0.61	0.95
		Boana (*B. raniceps*)	0.96	0.67	0.99
		Leptodactylus	0.88	0.65	0.96

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Table S5. Isometry tests for coefficients of the first eigenvector of pooled P-matrices and divergence matrices. The coefficients shown were obtained by dividing each original coefficient of the first eigenvector by the isometric expectation of 0.408. Coefficients in bold are significant because their 95% CI exclude one. Note that muscle CSA shows positive allometry, more strongly within species than between species.

Test for isometry	*Boana*			*Leptodactylus*		
Pooled P-matrices	empirical	CI 2.5%	CI 97.5%	empirical	CI 2.5%	CI 97.5%
femur	**0.798**	0.738	0.912	**0.867**	0.778	0.937
tibia	**0.812**	0.744	0.891	**0.823**	0.758	0.875
tarsus	**0.809**	0.740	0.918	**0.845**	0.765	0.909
foot	**0.909**	0.832	0.988	**0.752**	0.670	0.811
up. leg muscle CSA	**1.348**	1.215	1.421	**1.309**	1.225	1.425
low. leg muscle CSA	**1.198**	1.047	1.289	**1.257**	1.181	1.323
Divergence matrices	empirical	2.5%	97.5%	empirical	2.5%	97.5%
femur	**0.910**	0.858	0.960	0.943	0.880	1.007
tibia	**0.923**	0.851	0.994	**0.928**	0.870	0.984
tarsus	**0.831**	0.733	0.936	**0.917**	0.860	0.967
foot	0.985	0.930	1.042	**0.874**	0.802	0.937
up. leg muscle CSA	**1.143**	1.046	1.228	**1.136**	1.069	1.205
low. leg muscle CSA	**1.163**	1.070	1.250	**1.163**	1.069	1.261

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Table S6. First and second eigenvectors of the W-matrix for both frog genera. The W-matrix is a variance-covariance matrix of directional net-selection gradients on hindlimb traits, representing the realized adaptive landscape (AL). The eigenvector coefficients contributing the most to the major axes of the adaptive landscape are in bold. Eigenvalues indicate the strength of selection on each major axes of the AL.

Major axes AL	*Boana*		*Leptodactylus*	
	w_1	w_2	w_1	w_2
femur	0.050	0.075	**0.532**	-0.202
tibia	**-0.789**	**0.498**	**-0.790**	0.196
tarsus	-0.184	**-0.714**	0.029	0.141
foot	**0.578**	**0.463**	-0.295	**-0.835**
up. leg muscle CSA	0.056	-0.021	0.041	-0.165
low. leg muscle CSA	-0.058	0.144	0.045	**0.419**
eigenvalues	23.12	17.98	26.33	18.05

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Table S7. Vector correlations between eigenvectors of the W-matrix with eigenvectors of the pooled P-matrix and of the divergence matrix. Major axes of the AL are the first and second eigenvectors of the W-matrix (w_{max} and w_2). p_{max} and p_2 are the axes of most phenotypic variation, and d_{max} and d_2 are the axes of most phenotypic divergence. CI 2.5% and CI 97.5% are the bounds for confidence intervals for the correlations when accounting for uncertainty in all three matrices using bootstrapping. None of the correlations were significant because none was above the expected value for the null-distribution correlation between random vectors (0.76).

AL and pooled P-matrix							
w_{max} x p_{max}	empirical	CI 2.5%	CI 97.5%	w_{max} x p_2	empirical	CI 2.5%	CI 97.5%
Boana	0.09	0.00	0.16	*Boana*	0.14	0.02	0.20
Leptodactylus	0.10	0.04	0.14	*Leptodactylus*	0.08	0.05	0.26
w_2 x p_{max}				w_2 x p_2			
Boana	0.18	0.00	0.22	*Boana*	0.03	0.00	0.20
Leptodactylus	0.15	0.00	0.13	*Leptodactylus*	0.23	0.02	0.32
AL and divergence matrix							
w_{max} x d_{max}	empirical	CI 2.5%	CI 97.5%	w_{max} x d_2	empirical	CI 2.5%	CI 97.5%
Boana	0.11	0.05	0.18	*Boana*	0.34	0.08	0.46
Leptodactylus	0.15	0.04	0.18	*Leptodactylus*	0.26	0.01	0.39
w_2 x d_{max}				w_2 x d_2			
Boana	0.22	0.02	0.26	*Boana*	0.16	0.00	0.20
Leptodactylus	0.13	0.01	0.19	*Leptodactylus*	0.49	0.02	0.32

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Table S8. Jumping performance gradients for all six hindlimb traits within and between species. Within-species performance gradients were calculated using a multiple linear regression. Between-species performance gradients were calculated using a phylogenetic multiple linear regression assuming Brownian Motion for residual error and taking intraspecific standard errors into account. 95% CI were calculated using parametric standard errors. Bold values are significant under $\alpha = 0.05$.

Within-species	*B. boans*			*L. mystaceus*		
Maximum acceleration	slope	CI 2.5%	CI 97.5%	slope	CI 2.5%	CI 97.5%
femur	-0.10	-3.12	2.91	-1.07	-2.31	0.17
tibia	1.59	-1.73	4.92	0.65	-1.39	2.69
tarsus	-0.26	-1.87	1.35	-0.25	-1.56	1.06
foot	**-2.31**	-4.40	-0.22	-0.50	-2.08	1.09
up. leg muscle CSA	0.58	-0.31	1.47	0.55	-0.04	1.13
low. leg muscle CSA	-0.04	-0.63	0.55	0.26	-0.51	1.03
Maximum velocity						
femur	-0.50	-3.00	2.00	0.18	-0.93	1.29
tibia	1.96	-0.80	4.72	0.24	-1.59	2.07
tarsus	-0.82	-2.16	0.51	0.54	-0.64	1.71
foot	-0.71	-2.45	1.02	**-1.50**	-2.92	-0.07
up. leg muscle CSA	0.18	-0.56	0.91	**0.63**	0.11	1.16
low. leg muscle CSA	0.11	-0.38	0.60	-0.08	-0.77	0.62
Between-species	*Boana*			*Leptodactylus*		
Maximum acceleration	slope	CI 2.5%	CI 97.5%	slope	CI 2.5%	CI 97.5%
femur	0.84	-8.99	12.35	-0.02	-3.39	3.36
tibia	0.09	-16.27	17.66	-1.97	-7.35	3.42
tarsus	-0.56	-12.73	11.03	1.78	-3.71	7.27
foot	-1.98	-10.26	6.26	-0.34	-3.85	3.16
up. leg muscle CSA	2.12	-4.04	5.65	0.90	-3.63	5.43
low. leg muscle CSA	-1.78	-4.86	3.70	-1.16	-5.99	3.67
Maximum velocity						
femur	0.44	-7.16	7.83	**-1.47**	-2.82	-0.12
tibia	4.02	-7.75	16.08	0.11	-1.90	2.12
tarsus	-3.89	-12.30	4.39	0.58	-1.47	2.63
foot	-0.28	-6.02	5.59	1.20	-0.33	2.73
up. leg muscle CSA	0.24	-3.22	3.58	0.02	-1.79	1.83
low. leg muscle CSA	-0.61	-3.59	2.42	-0.39	-2.27	1.49

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Table S9. Jumping performance gradients when using total leg length and muscle CSA within and between species. Within-species performance gradients were calculated using a multiple linear regression. Between-species performance gradients were calculated using a phylogenetic multiple linear regression assuming Brownian Motion for residual error and taking intraspecific standard errors into account. 95% CI were calculated using parametric standard errors. Bold values are significant under $\alpha = 0.05$, and italicized values are significant under $\alpha = 0.10$.

Within-species	*Boana boans*			*Leptodactylus mystaceus*		
Maximum acceleration	slope	CI 2.5%	CI 97.5%	slope	CI 2.5%	CI 97.5%
leg length	*-0.89*	-1.79	0.01	**-1.23**	-2.25	-0.23
muscle CSA	0.26	-0.31	0.83	**0.84**	0.33	1.35
Maximum velocity						
leg length	-0.08	-0.80	0.64	-0.53	-1.47	0.40
muscle CSA	0.23	-0.23	0.68	**0.70**	0.23	1.17
Between-species	*Boana*			*Leptodactylus*		
Maximum acceleration	slope	CI 2.5%	CI 97.5%	slope	CI 2.5%	CI 97.5%
leg length	-0.54	-1.73	0.65	**-0.63**	-1.03	-0.23
muscle CSA	0.18	-1.04	1.40	-0.17	-0.57	0.23
Maximum velocity						
leg length	0.17	-1.06	1.39	0.37	-0.04	0.79
muscle CSA	-0.07	-1.29	1.14	**-0.51**	-0.92	-0.09

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Table S10. Vector correlations between performance gradients within- and between-species with the major axes of the AL and axes of pooled P-matrices. Performance gradients are partial regression coefficients of jumping performance on morphology. Major axes of the AL are the first two major axes of a variance-covariance matrix of directional selection (w_{max} and w_2 of the W-matrix). P_{max} and p_2 are the first two eigenvectors of the pooled P-matrix. Max acc = maximum jumping acceleration; max vel = maximum jumping velocity. The 5% upper tail of a null distribution of vector correlations was 0.76. 95% CI were obtained using bootstrapped pooled P-matrices and W-matrices.

Within-species								
AL and performance					Pooled P and performance			
Max acc	w_{max}	95% CI	w_2	95% CI	p_{max}	95% CI	p_2	95% CI
B. boans	0.87	0.23, 0.90	0.04	0.02, 0.84	0.05	0.01, 0.07	0.18	0.11, 0.24
L. mystaceus	0.61	0.47, 0.81	0.50	0.02, 0.66	0.02	0.002, 0.07	0.58	0.54, 0.62
Max vel	w_{max}		w_2		p_{max}		p_2	
B. boans	0.79	0.21, 0.88	0.52	0.13, 0.88	0.04	0.02, 0.06	0.07	0.03, 0.10
L. mystaceus	0.22	0.06, 0.65	0.69	0.16, 0.80	0.10	0.07, 0.13	0.21	0.12, 0.27
Between-species								
Max acc	w_{max}	95% CI	w_2		p_{max}	95% CI	p_2	
Boana	0.30	0.04, 0.51	0.43	0.01, 0.57	0.02	0.001, 0.06	0.01	0.003, 0.19
Leptodactylus	0.55	0.39, 0.65	0.16	0.07, 0.43	0.09	0.07, 0.12	0.06	0.005, 0.15
Max vel	w_{max}		w_2		p_{max}		p_2	
Boana	0.45	0.13, 0.80	0.82	0.16, 0.90	0.01	0.00, 0.04	0.06	0.01, 0.11
Leptodactylus	0.61	0.15, 0.72	0.37	0.05, 0.73	0.05	0.04, 0.09	0.04	0.002, 0.09

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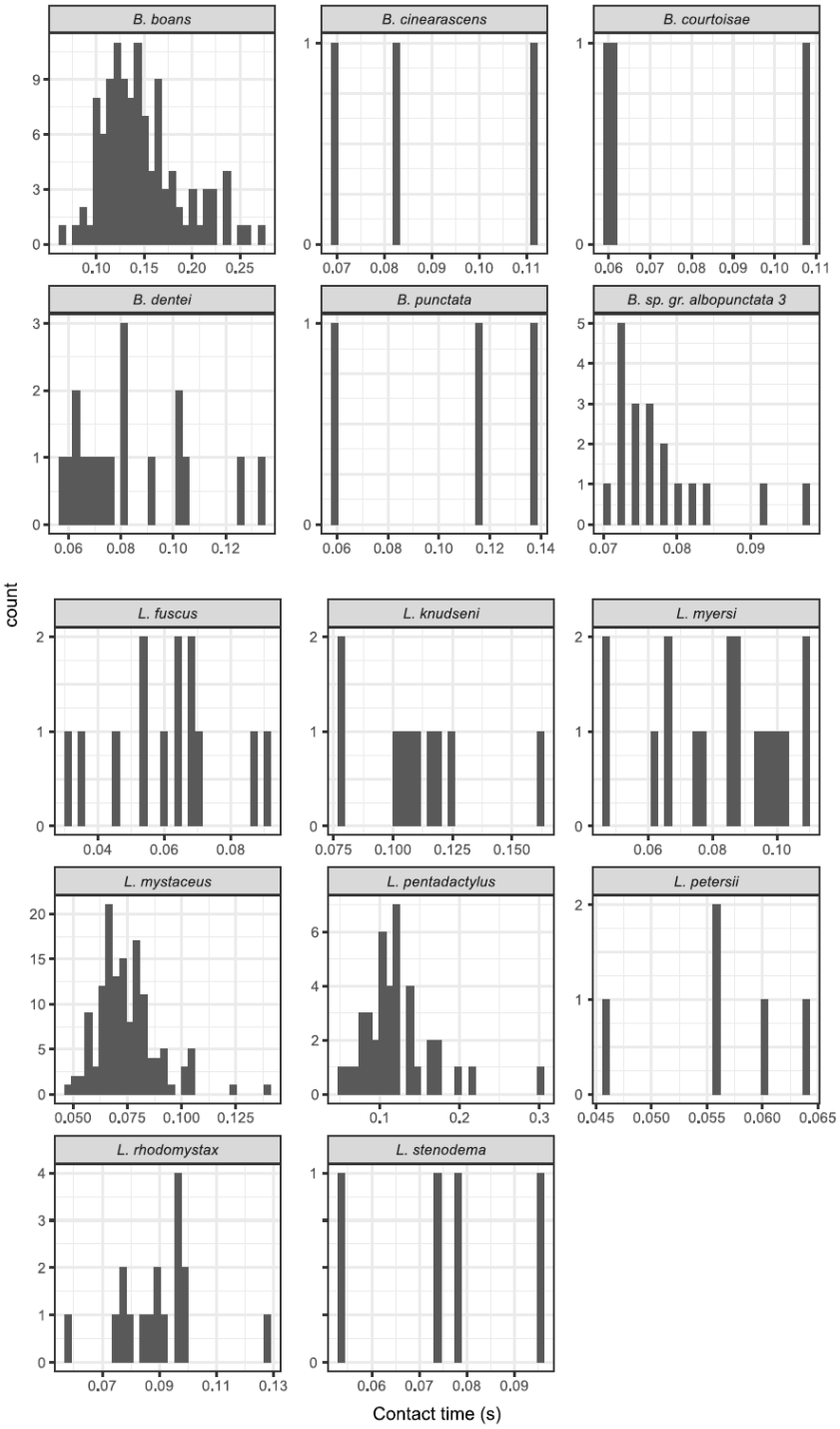

Figure S1. Final distribution of contact times across individuals within each species. We adopted the most stringent cut-off (i.e., 50%) to remove unrealistically high contact times from acceleration curves that resulted in exceptionally high velocity values (13 curves of 427 total).

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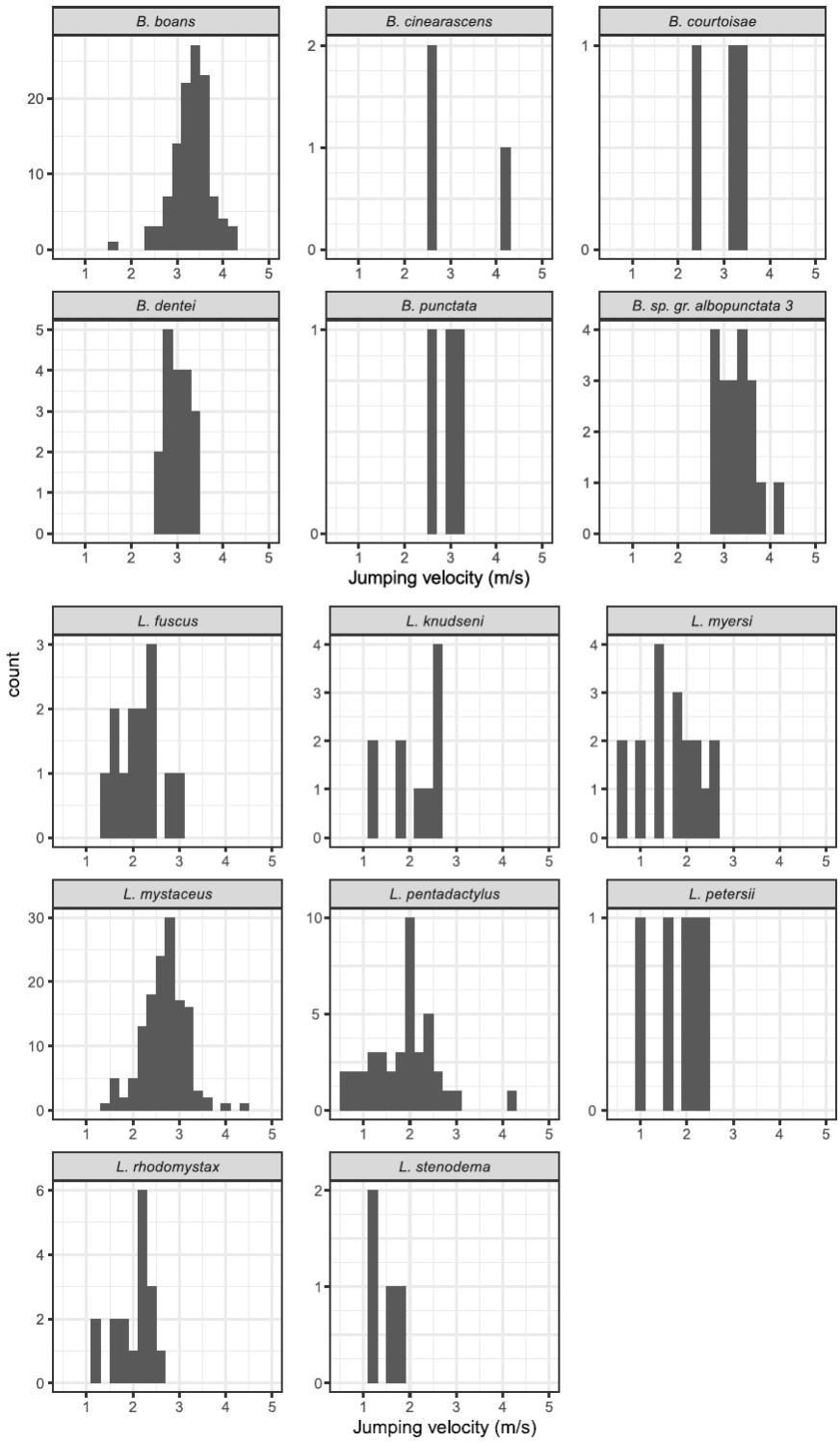

Figure S2. Final distribution of maximum velocities across individuals within each species.

Final values when adopting the most stringent cut-off of contact times as shown in Figure S1.

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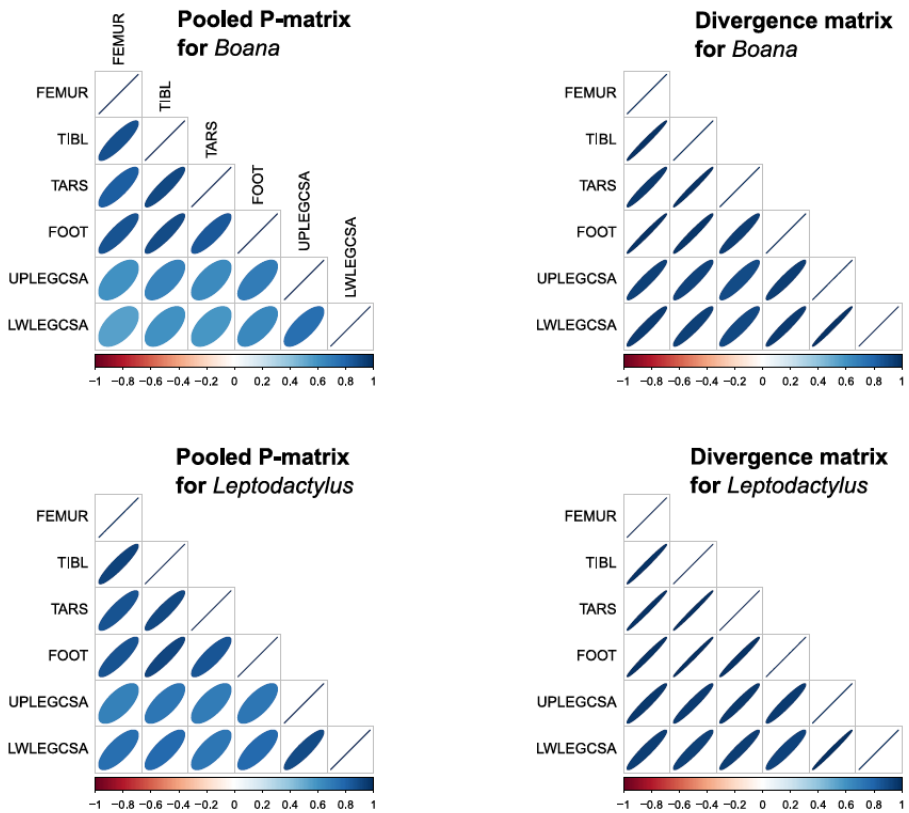

Figure S3. Pooled P-matrices and divergence matrices for *Boana* and *Leptodactylus*. The ellipses are 95% CI for the raw correlations among hindlimb traits. More circular ellipses, of lighter color, indicate lower correlations. FEMUR = femur length; TIBL = tibia length; TARS = tarsus length; FOOT = foot length; UPLEGCSA = upper leg muscle CSA; LWLEGCSA = lower leg muscle CSA. All matrices show the same pattern of higher correlations among variables that represent bone lengths or those that represent muscle CSA, contrasting with lower correlations between bone variables and muscle CSA variables.

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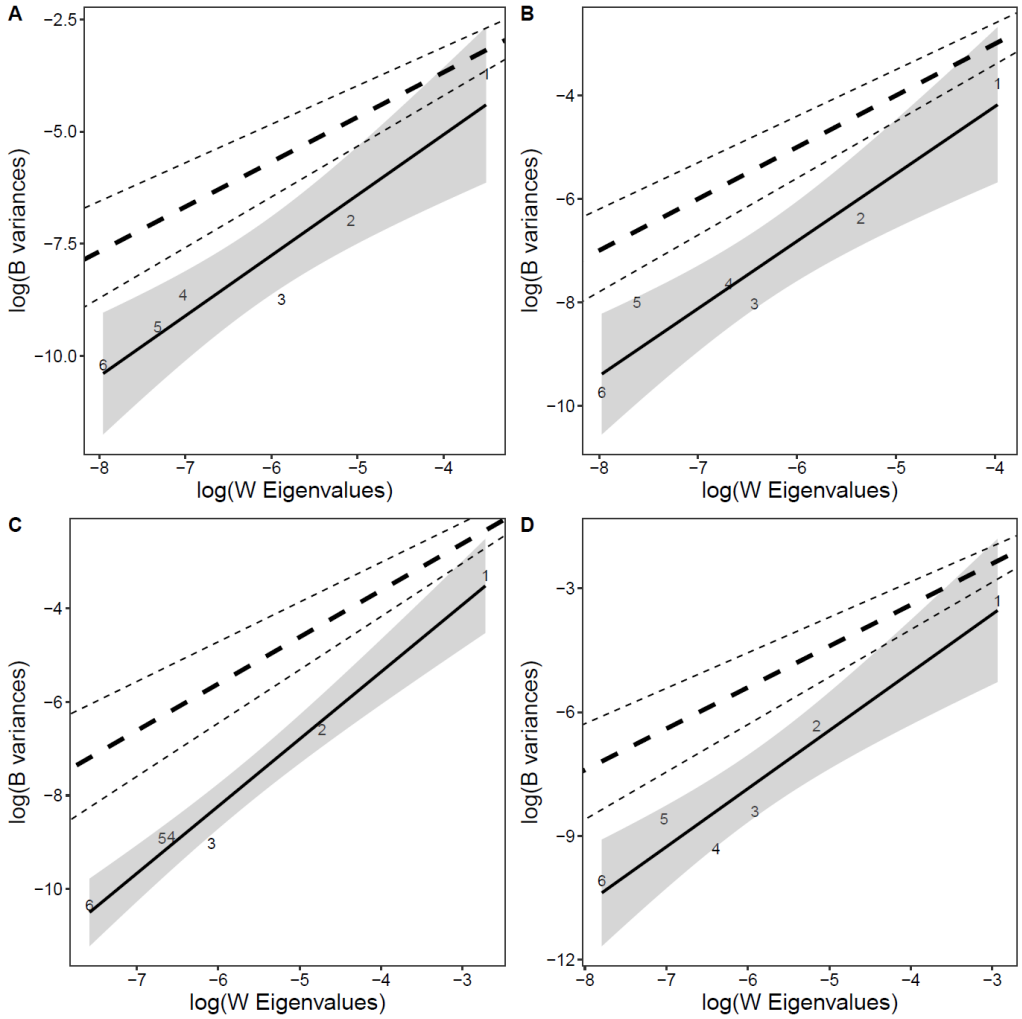

Figure S4. Regression plots for random drift tests in *Boana* (A, B) and *Leptodactylus* (C, D). The drift test regresses the variance between species (i.e., the variance of species' scores projected on eigenvectors of the pooled P-matrices) on within-species variation (i.e., **A, C:** pooled P-matrix eigenvalues, or **B, D**: *B. boans* P-matrix and *L. mystaceus* P-matrix, respectively). Numbers indicate eigenvalues for each eigenvector. The solid line is the empirical regression line, whereas the thick dashed line is the drift expectation of a slope equal to one, indicating variance between species proportional to variance within species. Thin dashed lines are 95% CI when performing BM simulations.

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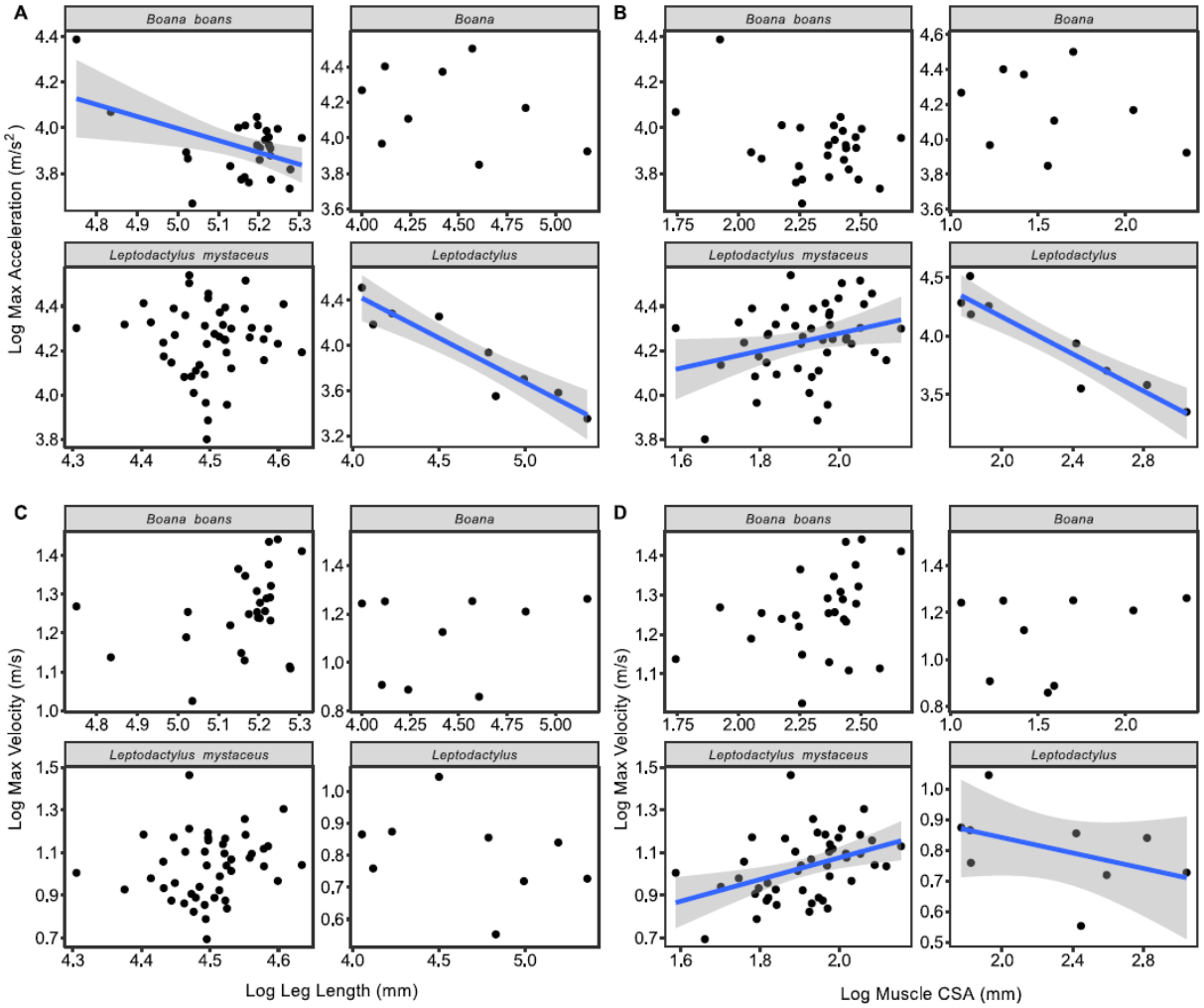

Figure S5. Relationships among jumping performance variables with total leg length and muscle CSA. All panels show the relationships at both within- and between-species levels that were tested using multiple linear regressions. We used ordinary least-squares for within-species regressions and phylogenetic generalized least-squares (assuming Brownian Motion) for between-species data. **A)** Maximum acceleration on leg length. **B)** Maximum acceleration on muscle CSA. **C)** Maximum velocity on leg length. **D)** Maximum velocity on muscle CSA. For maximum jumping acceleration, the proportion of variation explained by leg length and muscle CSA was: *B. boans*: R^2 = 0.22, *L. mystaceus*: R^2 = 0.16, *Leptodactylus*: R^2 = 0.97. For maximum

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jumping velocity, the proportion of variation explained by leg length and muscle CSA was: *L. mystaceus*: $R^2 = 0.16$, *Leptodactylus*: $R^2 = 0.57$.